MILITARY SPECIFICATION

COMPUTER, ATTACK
TYPE CPU-73 A/P

1. SCOPE

1.1 This specification covers one type of hand held attack computer.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitations for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

L-P-375	Plastic Film, Flexible, Vinyl Chloride
L-P-378	Plastic Sheet and Strip (Polyolefin)
L-P-535	Plastic Sheet (Sheeting); Vinyl Chloride Polymer and Vinyl Chloride - Vinyl Acetate Copolymer, Rigid
PPP-B-676	Boxes, Setup

Military

MIL-P-116	Preservation, Methods of
MIL-L-25142	Luminescent Material, Fluorescent
MIL-M-38733	Manuals Technical: Operation Instructions (For Hand-Held Flight Computers)

STANDARDS

Military

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-831	Test Reports, Preparation of
MS63033	Eyelets, Metallic, Rolled Deep Flange, with Washer, Style A

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Federal

FED-STD-595 Colors

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article inspection. This specification makes provisions for first article inspection.

3.2 Selection of specifications and standards. Specification and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3.1 Disks. The computer disks shall be constructed of vinyl chloride in accordance with Composition B, Type II, Grade C of L-P-535. The material shall be flat, homogeneous, and resistant to water and warpage.

3.3.2 Fastener. The fastener shall be a metal eyelet in accordance with MS63033 or other similar device.

3.3.3 Protective treatment. Use of materials that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage shall be avoided. When used they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.3.4 Case. The computer case material shall be .030 inch thick, in accordance with type II class 2 of L-P-375. The case dimensions shall be of sufficient size to adequately contain the computer and the operational manual described in 3.3.5. The case shall contain two (2) pockets. The case shall be marked or provide means for content identification. It shall be possible to close the case by the use of two (2) snaps or similar device(s). The following wording shall be printed on the front of the case:

"WARNING: Prolonged exposure to intense sunlight in a non-air conditioned enclosure may cause warpage."

"U. S. Government"

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The above wording shall conform to MIL-STD-130. The case color shall be #25102 of table VII of Fed-Std-595 or a commercial match.

3.3.5 Operational manual. One copy of the current operational manual shall be packed in the pocket of the case (ref para 3.3.4). The contractor will prepare the manual, if required by the contract, in accordance with the latest issue of MIL-M-38733. If preparation of a manual is required, the contractor will deliver the negatives to the government and request the necessary quantity of printed copies for packup. If the contract does not require preparation of an operational manual, the contractor will request the government to supply current printed copies for packup.

3.4 Design and construction. The computer shall consist of three rotatable discs mounted concentrically. The printed data shall be protected by an 0.005 inch thick clear vinyl, laminated under heat and pressure to form a protective bond. The rotating discs shall rotate freely and smoothly with no binding. The rotating parts shall be close fitted on the attachment shaft so that there will be no radial or axial play between them.

3.4.1 Printed data. Printed data contained on the computer shall be black over an opaque luminescent background, except that the data described in paragraph 3.4.4 shall not have an opaque luminescent background. The color of the background shall be in accordance with MIL-L-25142. Arrangement and location of tables, scales, and markings on the computer shall conform to figures 1 through 4 and shall be in accordance with 3.4.2 through 3.4.4.

3.4.2 Front disc. The front disc shall contain scales, tables, and other markings to allow the solution of time, speed, and distance problems. Specifically the front disc shall contain logarithmetic time and interceptor true air speed scales, a temperature-altitude window, a line of position (LOP) angle window, a lead distance window, double turn factor table, and a compressibility factor table. In addition, the front disc shall contain special markings as described in 3.4.2.5. The arrangement and location of the above scales, tables, and markings shall conform to Figure 1 and shall be in accordance with 3.4.2.1 through 3.4.2.11.

3.4.2.1 Dimensions. The outer radius of the front disc shall be 3.875 ± 0.025 inches, with the exception of an 0.500 ± 0.025 inch "spike" extending from the outer circumference of the 3.875 inch radius. The spike shall be approximately 0.5 inches wide at the base and shall taper to be tangent to an 0.125 inch diameter circle at the end. An 0.125 inch radius shall be used to blend the base of the "spike" to the outer circumference of the disc. The "spike" shall be centered about the index like described in 3.4.2.2.

3.4.2.2 Index line. A black index or "hairline" shall be contained on the front disc and shall provide a reference mark for all scale graduations

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on the front disc. The index line shall be approximately 0.005 inches wide and shall extend from the center of the disc through the "LOP window", terminate, then continue through the "Lead Distance Window", terminate and then continue again from the outer edge of the "Interceptor TAS" scale to the outer extremity of the disc.

3.4.2.3 Scale background. With the exception of the LOP, Lead Distance, and temperature-altitude windows, the front disc shall contain an opaque fluorescent background color in accordance with MIL-L-25142 for the area contained within a radius of 3.625 inches from the center of the disc. The LOP and temperature-altitude and lead distance windows, as well as all front disc material outside a radius of 3.625 inches from the center of the disc shall be of a clear transparent material in accordance with Federal Specification L-P-535, Composition B, Type II, Grade C.

3.4.2.4 Interceptor true airspeed scale. The interceptor true airspeed scale shall consist of logarithmic scale graduations for the purpose of providing solutions of time, speed, and distance problems. The location and arrangement of this scale shall be in accordance with Figure 1 and as defined in 3.4.2.4.1 through 3.4.2.4.4.

3.4.2.4.1 Interceptor true airspeed scale location and accuracy. The interceptor true airspeed scale shall begin and terminate at the index line defined in 3.4.2.2. The location of any graduation on this scale shall be defined by the equation $\theta = 360 \text{ Log } N/10$; where θ is equal to the number of degrees in a clockwise direction from the index line of 3.4.2.2 and N is the numerical value of the graduation. Acceptable nominal values for the location of graduation on this scale shall be within $\pm$10 minutes of arc of the nominal values listed in Table I.

TABLE I.
INTERCEPTOR AND TARGET TAS SCALES

Scale Graduation	Angle	
	Degrees	Minutes
10	0	0
10.1	1	33
10.2	3	6
10.3	4	37
10.4	6	8
10.5	7	38
10.6	9	7
10.7	10	35
10.8	12	2
10.9	13	28

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TABLE I. (Cont'd)

INTERCEPTOR AND TARGET TAS SCALES

Scale Graduation	Angle	
	Degrees	Minutes
11.0	14	54
11.1	16	19
11.2	17	43
11.3	19	6
11.4	20	29
11.5	21	51
11.6	23	12
11.7	24	33
11.8	25	53
11.9	27	12
12.0	28	30
12.1	29	48
12.2	31	5
12.3	32	22
12.4	33	38
12.5	34	53
12.6	36	8
12.7	37	22
12.8	38	36
12.9	39	49
13.0	41	1
13.1	42	13
13.2	43	24
13.3	44	35
13.4	45	45
13.5	46	55
13.6	48	4
13.7	49	13
13.8	50	21
13.9	51	29
14.0	52	36
14.1	53	43
14.2	54	49
14.3	55	55
14.4	57	1
14.5	58	6

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TABLE I. (Cont'd)

INTERCEPTOR AND TARGET TAS SCALES

Scale Graduation	Angle	
	Degrees	Minutes
14.6	59	10
14.7	60	14
14.8	61	18
14.9	62	21
15.0	63	24
15.2	65	28
15.4	67	30
15.6	69	31
15.8	71	31
16.0	73	29
16.2	75	26
16.4	77	21
16.6	79	14
16.8	81	7
17.0	82	58
17.2	84	47
17.4	86	36
17.6	88	23
17.8	90	9
18.0	91	54
18.2	93	38
18.4	95	20
18.6	97	1
18.8	98	42
19.0	100	21
19.2	101	59
19.4	103	37
19.6	105	13
19.8	106	48
20.0	108	22
20.2	109	56
20.4	111	28
20.6	113	00
20.8	114	30
21.0	116	00
21.2	117	29
21.4	118	57
21.6	120	24
21.8	121	51
22.0	123	16

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TABLE I. (Cont'd)
INTERCEPTOR AND TARGET TAS SCALES

Scale Graduation	Angle	
	Degrees	Minutes
22.2	124	41
22.4	126	5
22.6	127	29
22.8	128	51
23.0	130	13
23.2	131	35
23.4	132	55
23.6	134	15
23.8	135	34
24.0	136	53
24.2	138	10
24.4	139	28
24.6	140	44
24.8	142	0
25.0	143	16
25.2	144	30
25.4	145	44
25.6	146	58
25.8	148	11
26.0	149	23
26.2	150	35
26.4	151	47
26.6	152	57
26.8	154	8
27.0	155	17
27.2	156	27
27.4	157	35
27.6	158	44
27.8	159	51
28.0	160	59
28.2	162	5
28.4	163	12
28.6	164	18
28.8	165	23
29.0	166	28
29.2	167	32
29.4	168	36
29.6	169	40
29.8	170	43
30.0	171	46
30.5	174	21

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TABLE I. (Cont'd)
INTERCEPTOR AND TARGET TAS SCALES

Scale Graduation	Angle	
	Degrees	Minutes
31.0	176	53
31.5	179	24
32.0	181	51
32.5	184	17
33.0	186	40
33.5	189	01
34.0	191	20
34.5	193	37
35.0	195	52
35.5	198	05
36.0	200	16
36.5	202	26
37.0	204	33
37.5	206	39
38.0	208	43
38.5	210	46
39.0	212	47
39.5	214	46
40.0	216	44
40.5	218	41
41.0	220	36
41.5	222	30
42.0	224	22
42.5	226	13
43.0	228	03
43.5	229	51
44.0	231	39
44.5	233	25
45.0	235	09
45.5	236	53
46.0	238	36
46.5	240	17
47.0	241	57
47.5	243	37
48.0	245	15
48.5	246	52
49.0	248	28
49.5	250	03
50.0	251	38
50.5	253	11
51.0	254	44
51.5	256	15

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TABLE I. (Cont'd)
INTERCEPTOR AND TARGET TAS SCALES

Scale Graduation	Angle	
	Degrees	Minutes
52.0	257	46
52.5	259	15
53.0	260	44
53.5	262	12
54.0	263	40
54.5	265	06
55.0	266	32
55.5	267	57
56.0	269	21
56.5	270	44
57.0	272	07
57.5	273	29
58.0	274	50
58.5	276	11
59.0	277	30
59.5	278	50
60.0	280	08
61.0	282	43
62.0	285	16
63.0	287	46
64.0	290	13
65.0	292	39
66.0	295	02
67.0	297	23
68.0	299	42
69.0	301	59
70.0	304	14
71.0	306	27
72.0	308	38
73.0	310	48
74.0	312	55
75.0	315	01
76.0	317	06
77.0	319	08
78.0	321	09
79.0	323	09
80.0	325	07
81.0	327	03
82.0	328	58
83.0	330	52

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TABLE I. (Cont'd)
INTERCEPTOR AND TARGET TAS SCALES

Scale Graduation	Angle	
	Degrees	Minutes
84.0	332	44
85.0	334	35
86.0	336	25
87.0	338	14
88.0	340	01
89.0	341	47
90.0	343	32
91.0	345	15
92.0	346	58
93.0	348	39
94.0	350	20
95.0	351	59
96.0	353	37
97.0	355	14
98.0	356	50
99.0	358	26
100.0	360	00

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3.4.2.4.2 Interceptor true airspeed scale arrangement. The interceptor true airspeed scale markings shall be arranged as shown in figure 1 with initial number "10" outlined in black around, the background color for ease of recognization. The black marking surrounding the initial number shall blend into the index line defined in 3.4.2.2. All markings related to the interceptor true airspeed scale shall be contained within an 0.375 $\pm$ 0.025 inch band terminating at 3.025 $\pm$ 0.015 inches from the center of the disc.

3.4.2.4.3 Interceptor true airspeed scale designation. The interceptor true airspeed scale shall be designated by the words "INTERCEPTOR", "TAS", and "EAS" in 7 point, bold type, Gothic letters of an appropriate style (See Figure 1). Numerical values shall be located adjacent to the appropriate graduation for each whole number from 10 to 25 for every 5th number from 25 to 60 and every 10th number from 60 to 90. The size and style of numerical markings shall be approximately 8 1/2 point, bold face, Gothic. With the exception of the special markings specified in 3.4.2.5, scale graduation marks of approximately 0.15 inches in length shall be located at; each scale division of 0.5 between "10" and "15", each scale division of 1.0 between "15" and "60" and each scale division of 5.0 between "60" and "100". In addition, scale graduation marks of approximately 0.10 inches in length shall be located at each intervening scale division of; 0.1 between "10" and "15", 0.2 between "15" and "30", 0.5 between "30" and "60" and 1.0 between "60" and "100". The width of these graduation marks shall be approximately 0.005 inches.

3.4.2.4.4 Time scale. The words "Minutes" and "Seconds" shall be located within the boundary of the interceptor true airspeed scale to provide a reminder that the solution of problems involving time may be obtained by using the interceptor true airspeed scale. These words shall be located as shown in Figure 1 and shall be of 7 point, light, Gothic letters of an appropriate style.

3.4.2.5 Special markings. Special markings as described in 3.4.2.5.1 through 3.4.2.5.4 shall be included on the front disc.

3.4.2.5.1 Stern attack lead/lag index marks. Index marks, in the form of "arrow heads" shall be located at the appropriate position within the interceptor true airspeed scale band to aid in the solution of various stern attack problems. These "arrow heads" shall be approximately 0.150 inches long and 0.050 inches wide at the base with the arrow point terminating at the circumference of the interceptor true airspeed outer band. The designation of these index marks shall be located, where partical, under each index and shall be of a size as to be clearly discernible and yet not interfere with other markings on the disc. The location and designation of these markings are as follows: 90°S, 1.57 on interceptor TAS scale; 60°S, 1.05 on interceptor TAS scale; 130°S, 2.26 on interceptor TAS scale; 180°S, 3.14 on interceptor TAS scale.

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3.4.2.5.2 Time scale index mark. A special time scale index mark shall be located at the interceptor TAS scale mark for the number "60". This special marking shall consist of the number "60" outlined in black (see figure 1). The black outlining shall be bounded by a radial arc approximately 3.00 inches from the center of the disc and two radial arcs determined by the tangent points of two 0.75 inch radius circles tangent at the circumference of the interceptor TAS scale band and to the 3.00 inch radius.

3.4.2.5.3 20° Bank turn radius index. An index mark, in the form of an "arrow head", with the point terminating at 15.9 on circumference of the outer band of the interceptor true airspeed scale shall be provided for converting the turn radius required for a 30° bank angle to the turn radius required for a 20° bank angle. This arrow head shall be of such a size as to be clearly discernible and yet not interfere with the special marking for a 90° stern attack specified in 3.4.2.5.1. The designation of this mark shall be in the form of the number "20" contained within a lined rectangle below the interceptor true airspeed scale band (see figure 1). A line approximately 0.005 inches wide shall extend from the center of this rectangle to the base of the arrow head index mark.

3.4.2.5.4 60° Bank turn radius index. An index mark, in the form of an "arrow head", with the point terminating at 33.3 on the circumference of the interceptor true airspeed scale outer band shall be provided for converting the turn radius required for a 30° bank angle to the turn radius required for a 60° bank angle. This arrow head shall be designated by the number "60" and arranged as specified in 3.4.2.5.3 for the 20° bank index with the exception that the designation shall be contained within the interceptor TAS scale band.

3.4.2.6 Temperature-altitude window. The front disc shall contain a temperature-altitude window, which when used in conjunction with the front face of the center disc, shall provide solutions of problems relating to compensation for variation of air density as a function of pressure altitude and temperature. This window shall conform to Figure 1 and as defined in 3.4.2.6.1 through 3.4.2.6.5.

3.4.2.6.1 Temperature-altitude window location and dimensions. The temperature-altitude window shall consist of a clear transparent window bounded by two (2) arcs and two (2) radial segments. The inner arc shall be located at a distance of 1.800 ± 0.010 inches from the center of the disc and the outer arc at a distance of 2.100 ± 0.010 inches from the center of the disc. The radial segments shall be located at approximately 55° and 100°, measured in a clockwise direction from the interceptor TAS index line. The boundary of the window shall be outlined with a black line approximately 0.005 inches wide.

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3.4.2.6.2 Temperature-altitude window graduation marks. The temperature-altitude window shall contain graduation marks and numerical identification for an altitude and a temperature scale. The altitude scale shall be located along the inner arc segment boundary defined in 3.4.2.6.1. Graduation markings for the altitude scale shall be located as defined in Table II for altitudes in thousands of feet from 0 to 35,000 feet in 5,000 feet increments. Graduation markings for the temperature scale shall be positioned as specified in Table III for temperatures in degrees centigrade from +40°C to -80°C in 5°C increments. All graduation markings for these scales shall be within ± 5 minutes of arc of the values listed in Tables II and III. Numerical designations shall be provided above each 10°C increment of temperature and below each 10,000 feet increment of altitude over the entire range of the scales. Six (6) point bold face type shall be used for this designation.

TABLE II

ALTITUDE SCALE - TEMPERATURE-ALTITUDE WINDOW

Altitude - (1000 Ft)	Graduation Mark Location - CW	
	Degrees	Minutes
0	67°	00'
5	69°	44'
10	72°	34'
15	75°	30'
20	78°	33'
25	81°	43'
30	85°	02'
35	88°	29'

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TABLE III

TEMPERATURE SCALE, TEMPERATURE-ALTITUDE WINDOW

Temperature °C	Angle	
	Degrees	Minutes
+40	60	30
+35	61	45
+30	63	02
+25	64	20
+20	65	39
+15	67	00
+10	68	22
+ 5	69	46
0	71	11
- 5	72	37
-10	74	06
-15	75	36
-20	77	07
-25	78	41
-30	80	16
-35	81	54
-40	83	33
-45	85	15
-50	86	59
-55	88	45
-60	90	34
-65	92	25
-70	94	19
-75	96	16
-80	98	16

3.4.2.6.3 Temperature-altitude window designation. The letters "TEMP" and "ALT" in 7 point, bold face Gothic letters of an appropriate style shall be used to designate the temperature and altitude scales respectively. The location of these designations shall be as shown in figure 1.

3.4.2.6.4 Temperature-altitude window special markings. The words "35 to 80 STD DAY" shall be included for the altitude scale along with a special marking to designate that the graduation mark for 35,000 feet is to be used for computations involving altitudes between these limits. This special marking shall consist of a radial line terminating with an "arrow head" at the graduation mark for 35,000 feet and a circular arc underscoring the words "35 to 80 STD DAY". The designation "°C" shall be included to the right of the temperature scale to designate the unit of temperature. Six point, bold face type of an appropriate style shall be used for these designations.

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3.4.2.6.5 Temperature-altitude window graduation dimensions. Graduation markings for both the altitude and temperature scales shall be approximately 0.080 inches in length for each "5 increment mark" and 0.120 inches in length for each "10 increment mark" of both scales. The width of graduation marks shall be approximately 0.005 inches wide.

3.4.2.7 LOP window. The front disc shall contain a line of position (LOP) window for providing solutions of line of position problems for beam approach angles from 60 to 170 in 10 increments. The LOP window shall conform to figure 1 and as defined in 3.4.2.7.1 through 3.4.2.7.3.

3.4.2.7.1 LOP window location and dimensions. The LOP window shall consist of a clear transparent window bounded by the intersection of two (2) arcs and two radial lines. The inner arc boundary shall be located at a distance of approximately 0.350 inches from the center of the disc and the outer arc at a distance of 1.750 ±0.010 inches from the center of the disc. The radial lines which define the radial boundary of the window shall form an angle of approximately 17°from a point one (1) inch below the center of the disc. The radial lines shall begin at the inner arc boundary and terminate at the outer arc boundary. The boundary of the window shall be outlined with a black line approximately 0.005 inches wide. The LOP window shall be centered about the Interceptor TAS index line.

3.4.2.7.2 LOP window arrangement. The LOP window shall be sub-divided into twelve (12) equal spaces by arc segments extending away from each radial boundary of the window approximately 0.10 inches. Each of these spaces shall contain an appropriate numerical designation for beam approach angles from 60° to 170° in 10° increments. The numerical designation for 60° shall be located adjacent to the space the farthest from the center of the disc and the designation for 170° shall be located adjacent to the space closest to the center of the disc. The numerical designation for all even 10° angular increments shall be positioned to the left of the window and the numerical designation for all odd 10° angular increments shall be positioned to the right of the window. Six point, bold face numerals shall be used to designate the beam approach angle.

3.4.2.7.3 LOP window designation. The LOP window shall be designated by the words "LOP ANGLE" in 9 to 9 1/2 point, bold face, Gothic type of an appropriate style. The words "LOP ANGLE" shall be placed above the window as shown in figure 1.

3.4.2.8 Lead distance window. The front disc shall contain a window for providing solutions of the "lead distance problem" for both single and double turn beam attacks from 90° and 110°. The lead distance window shall be in accordance with figure 1 and as defined in 3.4.2.8.1 through 3.4.2.8.2.

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3.4.2.8.1 Lead distance window location and dimensions. The lead distance window shall consist of a clear transparent window bounded by the intersection of two (2) arcs and two radial lines. The inner arc boundary shall be located at a distance of 2.150 $\pm$ 0.010 inches from the center of the disc and the outer arc boundary shall consist of the inner circular boundary of the interceptor true air speed scale. The radial lines which define the radial boundary of the window shall consist of an extension of the radial boundary of the LOP window defined in 3.4.2.7.1. The radial lines shall begin at the inner arc boundary and terminate at the outer arc boundary. The boundary shall be outlined with a black line approximately 0.005 inches wide. The lead distance window shall be centered about the Interceptor TAS index line.

3.4.2.8.2 Window designation. The lead distance window shall be subdivided into two (2) equal spaces by arc segments extending away from each radial boundary of the window approximately 0.30 inches. Each of the above spaces shall be further subdivided into three (3) equal spaces by arc segments extending away from each radial boundary of window approximately 0.20 inches. The upper one-half of the window shall be designated by "90° HCA" and the lower one-half of the window shall be designated by "110° HCA" as shown in figure 1. The subdivided spaces for 90° HCA shall be designated with the words FINAL LEAD 40° TURN, DTF, and SINGLE TURN. The subdivided spaces for 110° HCA shall be designated with the words FINAL LEAD 30° TURN, DTF, and SINGLE TURN. These designations shall be positioned to the right of the window in the appropriate space as shown in figure 1. These designations shall be in 7 point bold face, Gothic type of an appropriate style.

3.4.2.9 Compressibility correction factor table. The front disc of the computer shall contain a compressibility factor table to provide information necessary in the conversion of calibrated airspeed to true airspeed. This table shall be in accordance with figure 1 and as defined in 3.4.2.9.1 through 3.4.2.9.3.

3.4.2.9.1 Compressibility correction factor data. The compressibility correction factor table, located on the front disc of the computer shall contain correction factors for altitudes from 10,000 feet through 30,000 feet in 5,000 feet increments for calibrated airspeeds from 175 knots to 950 knots. Correction factors shall be provided for each 25 knot increment of calibrated air speed between 175 knots and 750 knots and for each 50 knots between 750 and 950 knots. The data to be included in this table is contained in Table IV.

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TABLE IV
COMPRESSIBILITY CORRECTION FACTOR DATA

PRESSURE ALTITUDE FEET	CALIBRATED AIR SPEED - KNOTS															
	175	200	225	250	275	300	325	350	375	400	425	450	475	500	525	550
10,000	1	1	1	2	2	3	4	5	6	7	8	10	12	14	16	18
15,000	1	2	3	3	4	6	7	8	10	12	14	16	19	21	24	27
20,000	2	3	4	5	7	8	10	12	14	17	20	23	27	30	34	37
25,000	2	4	5	7	9	11	14	17	20	24	28	32	36	40	44	48
30,000	4	5	7	9	12	15	18	22	27	32	37	42	46	50	54	58

PRESSURE ALTITUDE	575	600	625	650	675	700	725	750	800	850	900	950
10,000	19	21	23	24	25	26	27	27	27	27	26	25
15,000	30	32	34	36	37	38	39	39	39	38	37	35
20,000	40	43	46	47	48	49	49	49	49	48	46	--
25,000	51	54	56	57	58	58	59	59	58	56	--	--
30,000	61	64	66	67	68	68	68	68	66	--	--	--

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3.4.2.9.2 Dimensions and layout. The compressibility correction factor table shall be arranged and positioned as shown in figure 1. The boundary of the table shall consist of lines approximately 0.005 inches wide. The upper horizontal boundary shall be approximately one-half inch from the center of the disc and shall run perpendicular to the Interceptor TAS' index line. The length of the upper and intermediate boundaries shall be approximately 4.50 inches long, but shall not extend beyond the inner Interceptor TAS scale boundary. The intermediate boundary (CAS from 175 to 550 knots) shall be located approximately .850 inches from the upper boundary. The lower boundary shall be located approximately 1.55 inches from the upper boundary, and shall be approximately 3.5 inches in length. The "pressure altitude" column of the table shall be approximately one-half inch in width and a space of approximately one-quarter inch shall be allowed for each calibrated air speed designation and the corresponding correction factors.

3.4.2.9.3 Table designation. The table shall be designated as shown in figure 1 using the following approximate letter sizes of an approximate style.

a. "COMPRESSIBILITY CORRECTION FACTORS" in 7 point Gothic.

b. "PRESS ALT FEET: in 4 point Gothic.

c. "CALIBRATED AIR SPEED - KNOTS" in 6 point Gothic.

d. "USE CAS AND P.A. TO OBTAIN CORRECTION. SUBTRACT THIS FACTOR FROM CAS BEFORE CONVERTING TO TAS" in 5 point Gothic.

e. The numerical designation for values of pressure altitude, calibrated air speed, and correction factors shall be in 5 point bold face numerals.

3.4.2.10 Double turn factor table. The front disc shall contain a Double Turn Factor Table to provide a "Rule of Thumb" method of determining the total lead for the Double Turn. This table shall be in accordance with figure 1 and as defined in 3.4.2.10.1 through 3.4.2.10.3.

3.4.2.10.1 Double turn factor data. The Double Turn Factor Table located on the front disc of the computer shall contain DT+ factors for interceptor speeds from 350 knots through 900 knots in increments of 50 knots. The data to be included in this table is contained in table XXVIII.

3.4.2.10.2 Dimensions and layout. The Double Turn Factor Table shall be arranged and positioned as shown in figure 1. The boundary of the table shall consist of lines approximately 0.005 inches wide. The left vertical boundary shall be located approximately 0.625 inches to the right of the center of the disc, and shall run parallel to the Interceptor TAS index line defined in paragraph 3.4.2.2. The right vertical line shall be located approximately 0.625 inches to right of the left vertical line and shall be parallel to the Interceptor TAS index line

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defined in paragraph 3.4.2.2. The vertical lines defined above shall be approximately 2.10 inches long. The center line as shown in figure 1 shall be approximately 1.95 inches long. A space of approximately 0.15 inches by 0.3125 inches shall be allowed for each Interceptor True Airspeed and corresponding DT+ factor. This table shall in no way interfere with any other scale, table, window, or designation thereof.

3.4.2.10.3 Table designation for double turn factor table. The table shall be designated as shown in figure 1 using the following approximate letter sizes of an appropriate style.

a. "DT FACTOR" in 6 point Gothic.

b. "ITAS" in 6 point Gothic.

c. "DT+" in 6 point Gothic.

d. Numerical designation for values of ITAS and DT+ shall be in 5 point bold face numerals.

3.4.2.11 Computer name plate. The designation of the computer shall be in accordance with figure 1 and as defined in 3.4.2.11.1 through 3.4.2.11.3.

3.4.2.11.1 The words "COMPUTER, ATTACK" shall be positioned as shown in figure 1. These words shall be in 9 1/2 point, bold face, Gothic type of an appropriate style.

3.4.2.11.2 The words TYPE CPU-73A/P shall be positioned as shown in figure 1. These words shall be in 7 point, bold face, Gothic type of an appropriate style.

3.4.2.11.3 The manufacturer's name, part number and the correct Federal Stock Number of this item shall be located below the type number specified in 3.4.2.11.2. The type size and style shall be at the option of the manufacturer, except that this information shall not interfere with any of the scale graduations on the front disc.

3.4.3 Center disc. The center disc shall contain scales, tables and other markings to allow the solution of various problems. The center disc shall be of a white opaque material with background and scale overlays conforming to 3.4.1. The scales and markings contained on the center disc shall conform to figures 2 and 3 and as described in 3.4.3.1 and 3.4.3.2. The outer radius of the center disc shall be 4.500 $\pm$ 0.025 inches.

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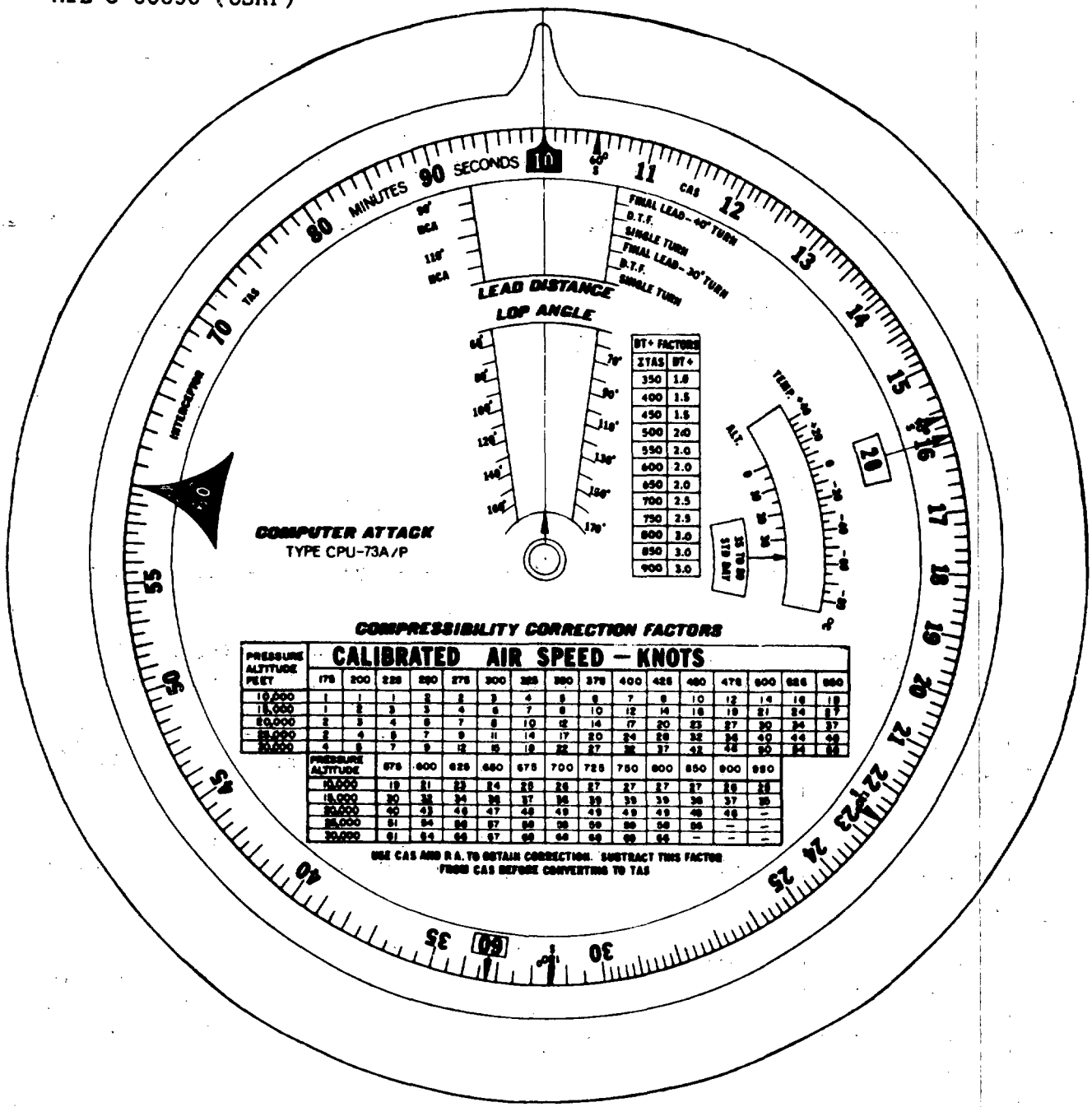


FIGURE I. FRONT DISC

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3.4.3.1 Center disc, front face. The front face of the center disc shall contain a Target True Airspeed (TAS) scale, twelve (12) Line of Position (LOP) scales, six (6) Lead Distance scales, a Pressure Altitude scale, two (2) Turn Radius scales, and various special markings. The front face of the center disc shall conform to figure 2 and as defined in 3.4.3.1.1 through 3.4.3.1.5.

3.4.3.1.1 Target true airspeed (TAS) scale. The target TAS scale shall contain graduation marks and scale designations as specified for the Interceptor TAS scale in 3.4.2.4.1 through 3.4.2.4.3. The location of these graduation marks shall be in accordance with Table I and as shown in figure 2. The target TAS scale shall be concentric and coincident with the Interceptor TAS scale. The target TAS scale shall contain special markings and wording as described in 3.4.3.1.1.1 through 3.4.3.1.1.5. The graduation markings shall extend radially from approximately 3.500 inches from the disc center to either 3.700 inches or 3.750 inches from the disc center for minor or major graduations respectively.

3.4.3.1.1.1 Index mark. The target TAS scale index mark shall be designated in a special manner for ease of recognition. This mark shall consist of the initial number "10" outlined in black around the background color. The index graduation shall provide a reference for all other graduation and markings on the front face of the center disc.

3.4.3.1.1.2 Test points. The target TAS scale shall contain three (3) special graduation markings which shall be used as test points for verification of scale accuracy. These special graduation markings shall be located at the graduation marks for the "index mark" (number 10) and numbers "20" and "50". These special graduation markings shall consist of a 0.020 inch wide graduation mark at each of the above locations. These markings shall extend from 3.5 inches from the disc center to 3.675 inches from the disc center. The special markings shall be centered about the actual graduation mark for the particular location.

3.4.3.1.1.3 Special designations. The Target TAS scale shall contain the words "TARGET TAS" as shown in figure 2 and in accordance with 3.4.2.4.3 for the Interceptor TAS scale. The Target TAS scale shall contain the word "SEC" in 7 point, medium faced, Gothic letters of an appropriate style. The work "SEC" shall be centered above the graduation mark for "36" on the Target TAS scale and an "arrow" shall extend from beneath the word and blend into the graduation mark. The letters "TAS" shall also be located on the target TAS scale, as shown in figure 2 (between numbers 45 and 50) and they shall be of the same size and style as "TARGET TAS" above.

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3.4.3.1.1.4 Interceptor travel indices. The target TAS scale shall contain two (2) special markings for ease of computation of lag/lead distance for 60° and 130° stern attack problems. These special markings shall be in the form of "arrow heads" located at the graduation marks for "87" and "77" respectively, as shown in figure 2. The designations for these marks shall be "TI", "60° S" and "TI", "130° S" as shown in figure 2. These designations shall be so located and of an appropriate size as not to interfere with other markings on the disc.

3.4.3.1.1.5 Percent computed turning point range. The front face of the center disc of the computer shall contain special markings to designate the percent of computed turning point range necessary to compensate for a deviation from a 90° stern attack. These markings shall be in the form of an "arrow head" as shown in figure 2 and shall be located at the nominal positions listed in Table V within ±20' of arc measured clockwise from the target TAS index. These marks shall be designated with the appropriate angle of stern attack in degrees and the letters SA to indicate that the angle is a stern attack angle. These designations shall be of 6 point alphabetic and numeric letter size of an appropriate style.

TABLE V. PERCENT COMPUTED TURNING POINT RANGE

SA	PERCENT	LOCATION	
		Degrees	Minutes
80°	82.6	330	11
70°	65.8	294	45
60°	50.0	251	39
50°	35.7	199	3
40°	23.4	132	55
30°	13.4	45	45

3.4.3.1.2 Line of position (LOP) scales. The front face of the center disc shall contain twelve (12) LOP scales which shall be used in conjunction with the "LOP ANGLE Window" described in 3.4.2.7. These scales shall contain graduation marks for the solution of line of position problems for beam approach angles from 60° to 170° in 10° increments. The graduation scales shall be located between approximately 1.750 and 0.350 inches from the center of the disc and such that the particular scale shall be coincident with the appropriate beam angle designation defined in 3.4.2.7.1 after assembly of the computer. The LOP scale conform to figure 2 and as defined in 3.4.3.1.2.1 through 3.4.3

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3.4.3.1.2.1 LOP scale graduation. Graduation marks shall be provided for each value of LOP specified in Tables VI through XVII for the particular scale. The angular location of each of these marks shall be within ±10 minutes of arc of the value specified in these tables. In addition these various scales shall be on closed concentric circles.

TABLE VI
60 BEAM APPROACH LOP SCALE LOCATION

LOP	LOCATION		LOP	LOCATION	
	DEGREES	MINUTES		DEGREES	MINUTES
30°	251°	38'	51°	331°	20'
31°	256°	17'	52°	334°	34'
32°	260°	49'	53°	337°	48'
33°	265°	13'	54°	341°	00'
34°	269°	30'	55°	344°	12'
35°	273°	31'	56°	347°	22'
36°	277°	47'	57°	350°	32'
37°	281°	47'	58°	353°	42'
38°	285°	41'	59°	356°	51'
39°	289°	32'	60°	0°	0'
40°	293°	18'	61°	3°	9'
41°	297°	0'	62°	6°	18'
42°	300°	38'	63°	9°	28'
43°	304°	13'	64°	12°	38'
44°	307°	45'	65°	15°	49'
45°	311°	14'	66°	19°	0'
46°	314°	40'	67°	22°	12'
47°	318°	4'	68°	25°	26'
48°	321°	26'	69°	28°	40'
49°	324°	46'	70°	31°	57'
50°	328°	3'			

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TABLE VII
70° BEAM APPROACH LOP SCALE LOCATION

LOP	LOCATION	
	DEGREES	MINUTES
26°	231°	55'
27°	237°	42'
28°	243°	19'
29°	248°	45'
30°	254°	1'
31°	259°	10'
32°	264°	10'
33°	269°	4'
34°	273°	50'
35°	278°	31'
36°	283°	6'
37°	287°	36'
38°	292°	1'
39°	296°	21'
40°	300°	38'
41°	304°	51'
42°	309°	0'
43°	313°	7'
44°	317°	11'
45°	321°	12'
46°	325°	11'
47°	329°	8'
48°	333°	3'

LOP	LOCATION	
	DEGREES	MINUTES
49°	336°	57'
50°	340°	49'
51°	344°	40'
52°	348°	31'
53°	352°	21'
54°	356°	11'
55°	0°	0'
56°	3°	49'
57°	7°	39'
58°	11°	29'
59°	15°	19'
60°	19°	11'
61°	23°	3'
62°	26°	57'
63°	30°	52'
64°	34°	49'
65°	38°	48'
66°	42°	50'
67°	46°	53'
68°	51°	0'
69°	55°	9'
70°	59°	22'

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TABLE VIII
80° BEAM APPROACH LOP SCALE LOCATION

LOP	LOCATION	
	DEGREES	MINUTES
24°	224°	4'
25°	230°	46'
26°	237°	14'
27°	243°	31'
28°	249°	38'
29°	255°	34'
30°	261°	21'
31°	266°	0'
32°	272°	32'
33°	277°	57'
34°	283°	16'
35°	288°	28'
36°	293°	36'
37°	298°	39'
38°	303°	38'
39°	308°	32'
40°	313°	24'
41°	318°	12'
42°	322°	57'
43°	327°	40'
44°	332°	21'
45°	336°	0'
46°	341°	38'
47°	346°	14'

LOP	LOCATION	
	DEGREES	MINUTES
48°	350°	50'
49°	355°	25'
50°	0°	0'
51°	4°	35'
52°	9°	10'
53°	13°	46'
54°	18°	22'
55°	23°	0'
56°	27°	39'
57°	32°	20'
58°	37°	3'
59°	41°	48'
60°	46°	36'
61°	51°	28'
62°	56°	22'
63°	61°	21'
64°	66°	24'
65°	71°	32'
66°	76°	45'
67°	82°	3'
68°	87°	28'
69°	93°	0'
70°	98°	39'

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TABLE IX

90° BEAM APPROACH LOP SCALE LOCATION

LOP	LOCATION	
	DEGREES	MINUTES
22°	218°	18'
23°	226°	2'
24°	233°	29'
25°	240°	43'
26°	247°	45'
27°	254°	35'
28°	261°	15'
29°	267°	45'
30°	274°	7'
31°	280°	22'
32°	286°	29'
33°	292°	30'
34°	298°	26'
35°	304°	17'
36°	310°	3'
37°	315°	46'
38°	321°	25'
39°	327°	1'
40°	332°	34'
41°	338°	6'
42°	343°	36'
43°	349°	5'
44°	354°	33'

LOP	LOCATION	
	DEGREES	MINUTES
45°	0°	0'
46°	5°	28'
47°	10°	56'
48°	16°	24'
49°	21°	54'
50°	27°	26'
51°	32°	59'
52°	38°	35'
53°	44°	14'
54°	49°	57'
55°	55°	43'
56°	61°	34'
57°	67°	30'
58°	73°	31'
59°	79°	39'
60°	85°	53'
61°	92°	15'
62°	98°	45'
63°	105°	25'
64°	112°	15'
65°	119°	17'
66°	126°	31'

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TABLE X

100° BEAM APPROACH LOP SCALE LOCATION

LOP	LOCATION	
	DEGREES	MINUTES
20°	214°	45'
21°	223°	39'
22°	232°	15'
23°	240°	35'
24°	248°	40'
25°	256°	32'
26°	264°	12'
27°	271°	42'
28°	279°	2'
29°	286°	14'
30°	293°	18'
31°	300°	16'
32°	307°	8'
33°	313°	55'
34°	320°	38'
35°	327°	17'
36°	333°	53'
37°	340°	27'
38°	346°	59'
39°	353°	30'
40°	0°	0'

LOP	LOCATION	
	DEGREES	MINUTES
41°	6°	30'
42°	13°	1'
43°	19°	33'
44°	26°	7'
45°	32°	43'
46°	39°	23'
47°	46°	5'
48°	52°	52'
49°	59°	44'
50°	66°	42'
51°	73°	47'
52°	80°	58'
53°	88°	19'
54°	95°	48'
55°	103°	28'
56°	111°	20'
57°	119°	25'
58°	127°	45'
59°	136°	21'
60°	145°	15'

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TABLE XI

110° BEAM APPROACH LOP SCALE LOCATION

LOP	LOCATION	
	DEGREES	MINUTES
18°	213°	39'
19°	223°	58'
20°	233°	56'
21°	243°	34'
22°	252°	54'
23°	261°	59'
24°	270°	51'
25°	279°	32'
26°	288°	2'
27°	296°	23'
28°	304°	36'
29°	312°	42'
30°	320°	44'
31°	328°	40'
32°	336°	33'
33°	344°	24'
34°	352°	12'
35°	0°	0'

LOP	LOCATION	
	DEGREES	MINUTES
36°	7°	48'
37°	15°	37'
38°	23°	27'
39°	31°	20'
40°	39°	17'
41°	47°	18'
42°	55°	24'
43°	63°	38'
44°	71°	59'
45°	80°	29'
46°	89°	9'
47°	98°	1'
48°	107°	6'
49°	116°	27'
50°	126°	5'
51°	136°	2'
52°	146°	22'

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TABLE XII
120° BEAM APPROACH LOP SCALE LOCATION

LOP	LOCATION	
	DEGREES	MINUTES
16°	215°	29'
17°	227°	35'
18°	239°	13'
19°	250°	27'
20°	261°	21'
21°	271°	58'
22°	282°	20'
23°	292°	28'
24°	302°	26'
25°	312°	15'
26°	321°	57'
27°	331°	32'
28°	341°	4'
29°	350°	33'
30°	0°	0'

LOP	LOCATION	
	DEGREES	MINUTES
31°	9°	28'
32°	18°	56'
33°	28°	28'
34°	38°	4'
35°	47°	45'
36°	57°	34'
37°	67°	32
38°	77°	41'
39°	88°	2'
40°	98°	39'
41°	109°	33'
42°	120°	48'
43°	132°	26'
44°	144°	31'

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TABLE XIII
130° BEAM APPROACH LOP SCALE LOCATION

LOP	LOCATION	
	DEGREES	MINUTES
14°	221°	12'
15°	235°	35'
16°	249°	24'
17°	262°	44'
18°	275°	41'
19°	288°	17'
20°	300°	38'
21°	312°	45'
22°	324°	43'
23°	336°	32'
24°	348°	18'
25°	0°	0'

LOP	LOCATION	
	DEGREES	MINUTES
26°	11°	43'
27°	23°	28'
28°	35°	18'
29°	47°	15'
30°	59°	22'
21°	71°	43'
32°	84°	20'
33°	97°	16'
34°	110°	36'
35°	124°	25'
36°	138°	48'

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TABLE XIV

140° BEAM APPROACH LOP SCALE LOCATION

LOP	LOCATION	
	DEGREES	MINUTES
12°	232°	40'
13°	250°	13'
14°	267°	4'
15°	283°	20'
16°	299°	10'
17°	314°	40'
18°	329°	55'
19°	345°	0'
20°	0°	0'

LOP	LOCATION	
	DEGREES	MINUTES
21°	15°	1'
22°	30°	6'
23°	45°	21'
24°	60°	50'
25°	76°	40'
26°	92°	57'
27°	109°	47'
28°	127°	21'

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TABLE XV

150° BEAM APPROACH LOP SCALE LOCATION

LOP	LOCATION	
	DEGREES	MINUTES
9°	230°	24'
10°	254°	2'
11°	276°	28'
12°	298°	3'
13°	319°	1'
14°	339°	36'

LOP	LOCATION	
	DEGREES	MINUTES
15°	0°	0'
16°	20°	24'
17°	40°	59'
18°	61°	58'
19°	83°	32'
20°	105°	59'

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TABLE XVI

160° BEAM APPROACH LOP SCALE LOCATION

LOP	LOCATION	
	DEGREES	MINUTES
6°	228°	48'
7°	264°	11'
8°	297°	15'
9°	328°	57'
10°	0°	0'
11°	31°	4'
12°	62°	46'
13°	95°	50'
14°	131°	13'

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TABLE XVII
170° BEAM APPROACH LOP SCALE

LOP	LOCATION	
	DEGREES	MINUTES
3°	227°	50'
4°	296°	45'
5°	0°	00'
6°	63°	14'
7°	132°	9'

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3.4.3.1.2.2 LOP scale designations. Numerical designations shall be provided for each even value of LOP for all LOP scales with the exception of the "170° Beam Approach" scale which shall contain designations for each value of LOP listed. These numerical designations shall be approximately 4 point medium face numerals of an appropriate style.

3.4.3.1.3 Pressure altitude scale. The front face of the center disc shall contain a pressure altitude scale for providing solutions to problems involving compensation for variation of air density with pressure and temperature. This scale will be used in conjunction with the temperature altitude window described in 3.4.2.6 and all graduations and markings on this scale shall be clearly visible through the temperature altitude window after the computer is assembled. The pressure altitude scale shall conform to figure 2 and as defined in 3.4.3.1.3.1 through 3.4.3.1.3.4.

3.4.3.1.3.1 Pressure altitude scale graduations. Scale graduation marks shall be provided at the appropriate location for pressure altitudes from -2000 feet to +75000 feet in 1000 feet increments. The angular location of these marks shall be within ±10 minutes of arc of the value specified in Table XVIII. These graduation marks shall be located radially from the center of the disc in such a manner as to extend into the upper and lower dimensional boundary of the temperature-altitude window approximately 0.075 inches after the computer is assembled. These graduation marks shall be approximately 0.005 inches in width.

TABLE XVIII
PRESSURE ALTITUDE SCALE

PRESSURE ALTITUDE IN 1,000 FEET	ANGLE		PRESSURE ALTITUDE IN 1,000 FEET	ANGLE	
	DEGREES	MINUTES		DEGREES	MINUTES
-2	61	23	6	84	19
-1	64	11	7	87	16
0	67	0	8	90	15
1	69	50	9	93	15
2	72	41	10	96	16
3	75	34	11	99	19
4	78	28	12	102	23
5	81	22	13	105	28

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TABLE XVIII, (cont'd)

PRESSURE ALTITUDE SCALE

PRESSURE ALTITUDE IN 1,000 FEET	ANGLE		PRESSURE ALTITUDE IN 1,000 FEET	ANGLE	
	DEGREES	MINUTES		DEGREES	MINUTES
14	108	35	39	195	7
15	111	43	40	198	52
16	114	53	41	202	38
17	118	4	42	206	23
18	121	17	43	210	9
19	124	31	44	213	54
20	127	47	45	217	40
21	131	4	46	221	25
22	134	23	47	225	10
23	137	44	48	228	56
24	141	6	49	232	41
25	144	30	50	236	27
26	147	55	51	240	12
27	151	22	52	243	58
28	154	51	53	247	43
29	158	22	54	251	28
30	161	55	55	255	14
31	165	29	56	258	59
32	169	6	57	262	45
33	172	44	58	266	30
34	176	24	59	270	16
35	180	6	60	274	1
36	183	51	61	277	47
37	187	36	62	281	32
38	191	21	63	285	17

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TABLE XVIII, (cont'd)
PRESSURE ALTITUDE SCALE

PRESSURE ALTITUDE IN 1,000 FEET	ANGLE		PRESSURE ALTITUDE IN 1,000 FEET	ANGLE	
	DEGREES	MINUTES		DEGREES	MINUTES
64	289	3	70	311	32
65	292	48	71	315	16
66	296	34	72	318	60
67	300	19	73	322	43
68	304	4	74	326	26
69	307	48	75	330	9

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3.4.3.1.3.2 Pressure altitude scale designation. The designation "PRESS ALT" shall be used to identify the pressure altitude scale. The letter size and style shall be as specified in 3.4.2.6.3 for the "TEMP" and "ALT" designations used for the temperature-altitude window. This designation shall be located in a space adjacent to and counter clockwise from the beginning of the scale graduation marks. This designation shall be centered about an arc passing through the center of the temperature-altitude window, upon assembly of the computer.

3.4.3.1.3.3 Pressure altitude scale graduation designations. Numerical scale designations shall be provided for each 5,000 foot increment of pressure altitude from 0 to 75,000 feet. The designation shall be in thousands of feet and centered within the temperature-altitude window upon assembly of the computer. The size and style of these numerals shall be the same as for the "PRESS ALT" designation specified in 3.4.3.1.3.2.

3.4.3.1.3.4. Mach index. The pressure altitude scale shall contain a special graduation mark, designated by the words "MACH INDEX". This graduation shall consist of two "arrow heads" connected at the bases by a 0.0005 inch thick line. The points of the arrow heads shall terminate in a line extending just beyond the boundaries of the Temperature-Altitude Window when the computer is assembled. This index shall be located at 2° 23' $\pm$ 10' clockwise from the Target TAS index line. The word "MACH" shall be located just clockwise of the graduation mark and the word "INDEX" shall be located just clockwise of the graduation mark as shown in figure 2. The size and style of this designation shall be the same as used for the pressure altitude scale designation.

3.4.3.1.4 Turn radius scales. The front face of the center disc shall contain two turn radius scales as shown in figure 2 and as defined in 3.4.3.1.4.1 and 3.4.3.1.4.2.

3.4.3.1.4.1 30° Bank angle turn radius scale. The computer shall contain a scale for obtaining the turn radius required for a 30° bank angle between speeds of 280 and 1700 knots. This scale shall be located on the arc of a circle approximately 4.15 inches from the center of the disc. The scale shall be identified by the words "TURN RADIUS (30°B)" as shown in figure 2. This scale shall contain graduation marks for each one-quarter nautical mile between 2 and 10 nautical miles and for each nautical mile for values greater than 10 nautical miles. Numerical designation shall be provided for each nautical mile between 2 and 10 nautical miles, each even nautical mile between 10 and 50 nautical miles and for every fifth nautical mile for values greater than 50 nautical miles. The graduation mark for each designated value shall be approximately 0.100 inches in length and the graduation mark for each undesignated value shall be approximately 0.075 inches in length. Each graduation mark shall be approximately 0.005 inches in width. The location of each graduation mark shall be referenced clockwise from the Target TAS index and shall be within $\pm$ 10 arc minutes of the values specified in Table XIX.

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TABLE XIX

30° BANK ANGLE TURN RADIUS
SCALE LOCATION

TURN RADIUS	LOCATION	
	DEGREES	MINUTES
2.00	161	46
2.25	170	58
2.50	179	12
2.75	186	39
3.00	193	27
3.25	199	43
3.50	205	30
3.75	210	54
4.00	215	57
4.25	220	41
4.50	225	9
4.75	229	23
5.00	233	23
5.25	237	12
5.50	240	50
5.75	244	19
6.00	247	38
6.25	250	50
6.50	253	54
6.75	256	51

TURN RADIUS	LOCATION	
	DEGREES	MINUTES
7.00	259	41
7.25	262	26
7.50	265	5
7.75	267	39
8.00	270	8
8.25	272	32
8.50	274	52
8.75	277	8
9.00	279	20
9.25	281	29
9.50	283	34
9.75	285	36
10.00	287	34
11.00	295	1
12.00	301	50
13.00	308	5
14.00	313	53
15.00	319	16
16.00	324	19
17.00	329	3

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TABLE XIX.

30° Bank Angle Turn Radius Scale Location.

TURN RADIUS	LOCATION	
	DEGREES	MINUTES
18.00	333	31
19.00	337	45
20.00	341	46
21.00	345	34
22.00	349	13
23.00	352	41
24.00	356	1
25.00	359	12
26.00	2	16
27.00	5	13
28.00	8	4
29.00	10	48
30.00	13	27
31.00	16	1
32.00	18	30
33.00	20	54
34.00	23	14
35.00	25	30
36.00	27	42
37.00	29	51
38.00	31	56
39.00	33	57
40.00	35	56
41.00	37	52
42.00	39	45

TURN RADIUS	LOCATION	
	DEGREES	MINUTES
43.00	41	35
44.00	43	23
45.00	45	9
46.00	46	52
47.00	48	33
48.00	50	11
49.00	51	48
50.00	53	23
51.00	54	56
52.00	56	27
53.00	57	56
54.00	59	24
55.00	60	50
56.00	62	14
57.00	63	37
58.00	64	59
59.00	66	19
60.00	67	38
61.00	68	55
62.00	70	12
63.00	71	27
64.00	72	41
65.00	73	53
66.00	75	5
67.00	76	16

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TABLE XIX (cont'd) 30° BANK
ANGLE TURN RADIUS
SCALE LOCATION

TURN RADIUS	LOCATION	
	DEGREES	MINUTES
68.00	77	25
69.00	78	34
70.00	79	41
71.00	80	48
72.00	81	53
73.00	82	58
74.00	84	2
75.00	85	5

3.4.3.1.4.2 45 Bank angle turn radius scale. The computer shall contain a scale for obtaining the turn radius required for a 45 angle between speeds of 370 and 2300 knots. This scale shall be located on the arc of a circle approximately 4.40 inches from the center of the disc. The scale shall be identified by the words "TURN RADIUS (45° B)" as shown in figure 2. This scale shall contain graduation marks for each one-quarter nautical mile between 2 and 10 nautical miles and for each nautical mile for values greater than 10 nautical miles. Numerical designations shall be provided for each nautical mile between 2 and 10 nautical miles, each even nautical mile between 10 and 50 nautical miles and for every fifth nautical mile for values greater than fifty nautical miles. The graduation mark for each designated value shall be approximately 0.100 inches in length and the graduation mark for each undesignated value shall be approximately 0.075 inches in length. Each graduation mark shall be approximately 0.005 inches in width. The location of each graduation mark shall be referenced clockwise from the Target TAS index and shall be within ±10 minutes of arc of the values specified in Table XX.

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TABLE XX
45° BANK ANGLE TURN RADIUS
SCALE LOCATION

TURN RADIUS	LOCATION	
	DEGREES	MINUTES
2.00	204	42
2.25	213	54
2.50	222	9
2.75	229	36
3.00	236	24
3.25	242	39
3.50	248	27
3.75	253	50
4.00	258	53
4.25	263	37
4.50	268	6
4.75	272	19
5.00	276	20
5.25	280	9
5.50	283	47
5.75	287	15
6.00	290	35
6.25	293	46
6.50	296	50
6.75	299	47

TURN RADIUS	LOCATION	
	DEGREES	MINUTES
7.00	302	38
7.25	305	23
7.50	308	2
7.75	310	35
8.00	313	4
8.25	315	29
8.50	317	49
8.75	320	5
9.00	322	17
9.25	324	25
9.50	326	30
9.75	328	32
10.00	330	31
11.00	337	58
12.00	344	46
13.00	351	1
14.00	356	49
15.00	2	13
16.00	7	15
17.00	12	0

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TABLE XX (Cont'd) 45°
ANGLE TURN RADIUS
SCALE LOCATION

TURN RADIUS	LOCATION		TURN RADIUS	LOCATION	
	DEGREES	MINUTES		DEGREES	MINUTES
18.00	16	28	36.00	70	38
19.00	20	41	37.00	72	47
20.00	24	42	38.00	74	52
21.00	28	30	39.00	76	54
22.00	32	9	40.00	78	53
23.00	35	37	41.00	80	48
24.00	38	57	42.00	82	41
25.00	42	8	43.00	84	32
26.00	45	12	44.00	86	20
27.00	48	9	45.00	88	5
28.00	51	0	46.00	89	48
29.00	53	44	47.00	91	29
30.00	56	23	48.00	93	8
31.00	58	57	49.00	94	45
32.00	61	26	50.00	96	19
33.00	63	50	51.00	97	52
34.00	66	10	52.00	99	23
35.00	68	26	53.00	100	53

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TABLE XX (Cont'd) 45°
ANGLE TURN RADIUS
SCALE LOCATION

TURN RADIUS	LOCATION		TURN RADIUS	LOCATION	
	DEGREES	MINUTES		DEGREES	MINUTES
54.00	102	20	72.00	124	50
55.00	103	46	73.00	125	54
56.00	105	11	74.00	126	58
57.00	106	34	75.00	128	1
58.00	107	55	76.00	129	3
59.00	109	16	77.00	130	4
60.00	110	34	78.00	131	5
61.00	111	52			
62.00	113	8			
63.00	114	23			
64.00	115	37			
65.00	116	50			
66.00	118	1			
67.00	119	12			
68.00	120	21			
69.00	121	30			
70.00	122	37			
71.00	123	44			

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3.4.3.1.5 Lead distance scales. The front face of the center computer disc shall contain six lead distance scales for use in conjunction with the lead distance window described in 3.4.2.8. These scales shall contain graduation marks for the solution of lead distance problems for single and double turn beam attacks from 90° and 110°. These scales shall be located between approximately 2.150 and 3.275 inches from the center of the disc and such that the particular scale shall be coincident with the appropriate scale scale designation defined in 3.4.2.8 after assembly of the computer. The lead distance scales shall conform to figure 2 and as defined in 3.4.8.1.5.1 through 3.4.3.1.5.6. The arcs upon which the various graduations are placed shall be closed concentric circles.

3.4.3.1.5.1 90° HCA, single turn scale. The 90° HCA SINGLE TURN scale shall be located within a circular band defined by the lower one-third of the 90° HCA section of the lead distance window as defined in 3.4.2.8 and shall be contained within the boundaries of the "SINGLE TURN" designation adjacent to this window following assembly of the computer. This scale shall contain graduation marks for each value of LOP angle between 22° and 66°. The location of these graduation marks shall be referenced clockwise from the Target TAS scale index and shall be positioned to within ± 10 arc minutes of the values specified in Table XXI. Numerical designations shall be provided for every fifth value of LOP between 25° and 60° and for every even value of LOP between 60° and 66°. These numerical designations shall be approximately 7 point, medium face numerals of an appropriate style.

TABLE XXI
90° HCA
SINGLE TURN LEAD SCALE LOCATION

LOP	LOCATION	
	DEGREES	MINUTES
22°	32°	27'
23°	33°	55'
24°	35°	24'
25°	36°	54'
26°	38°	24'
27°	39°	55'
28°	41°	26'
29°	42°	59'
30°	44°	32'

LOP	LOCATION	
	DEGREES	MINUTES
31°	46°	6'
32°	47°	6'
33°	49°	18'
34°	50°	55'
35°	52°	34'
36°	54°	14'
37°	55°	56'
38°	57°	39'
39°	59°	24'

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TABLE XXI
90° HCA
SINGLE TURN LEAD SCALE LOCATION

LOP	LOCATION	
	DEGREES	MINUTES
40°	61°	11'
41°	63°	0'
42°	64°	50'
43°	66°	43'
44°	68°	38'
45°	70°	36'
46°	72°	37'
47°	74°	40'
48°	76°	46'
49°	78°	55'
50°	81°	8'
51°	83°	51'
52°	85°	45'

LOP	LOCATION	
	DEGREES	MINUTES
53°	88°	10'
54°	90°	39'
55°	93°	13'
56°	95°	52'
57°	98°	37'
58°	101°	27'
59°	104°	25'
60°	107°	29'
61°	110°	41'
62°	114°	1'
63°	117°	30'
64°	121°	9'
65°	124°	58'
66°	129°	0'

3.4.3.1.5.2 90° HCA, double turn factor (DTF) scale. The 90° HCA DTF scale shall be located within a circular band defined by the center one-third of the 90° HCA section of the Lead Distance Window defined in 3.4.2.8 and shall be contained within the boundaries of the DTF designation adjacent to this window following assembly of the computer. This scale shall contain graduation marks for each value of LOP angle between 9° and 70°. The location of these graduation marks shall be referenced clockwise from the Target TAS scale index and shall be positioned within ± 10 arc minutes of the values specified in Table XXII. These graduation marks shall be approximately 0.005 inches wide and 0.050 inches in length. Numerical designations shall be approximately 7 point, medium face numerals of an appropriate style.

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TABLE XXII
90° HCA
DOUBLE TURN FACTOR SCALE LOCATION

LOP	LOCATION	
	DEGREES	MINUTES
9	90	54
10	75	5
11	60	48
12	47	47
13	35	50
14	24	46
15	14	29
16	4	51
17	355	48
18	347	16
19	339	11
20	331	30
21	324	10
22	317	10
23	310	28
24	304	1
25	297	49
26	291	50
27	286	3
28	280	27
29	275	1

LOP	LOCATION	
	DEGREES	MINUTES
30	269	44
31	264	36
32	259	35
33	254	42
34	249	55
35	245	14
36	240	39
37	236	10
38	231	45
39	227	24
40	223	7
41	218	55
42	214	45
43	210	38
44	206	35
45	202	33
46	198	34
47	194	37
48	190	42
49	186	48
50	182	56

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TABLE XXII (Cont'd)
90° HCA DOUBLE TURN FACTOR SCALE LOCATION

LOP	LOCATION	
	DEGREES	MINUTES
51	179	5
52	175	14
53	171	24
54	167	35
55	163	45
56	159	56
57	156	6
58	152	16
59	148	26
60	144	34
61	140	42
62	136	48
63	132	53
64	128	56
65	124	57
66	120	56
67	116	52
68	112	45
69	108	36
70	104	23

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3.4.3.1.5.3 90° HCA, final lead - 40° turn scale. The 90° HCA FINAL LEAD - 40° turn scale shall be located within a circular band defined by the upper one-third of the 90° HCA section of the lead distance window as defined in 3.4.2.8 and shall be contained within the boundaries of the "FINAL LEAD" - 40° turn designation adjacent to this window following assembly of the computer. This scale shall contain graduation marks for each value of LOP angle between 22° and 70°. The location of these graduation marks shall be referenced clockwise from the Target TAS scale index and shall be positioned within ± 10 arc minutes of the values specified in Table XXIII. These graduation marks shall be approximately 0.005 inches wide and 0.050 inches in length. Numerical designations shall be provided for every fifth value of LOP, between 25° and 60° and for every even value of LOP between 60° and 70°. These numerical designations shall be approximately 7 point, medium face numerals of an appropriate style.

TABLE XXIII
90° HCA DOUBLE TURN
FINAL TURN LEAD DISTANCE (DT-FLD) SCALE LOCATION

LOP	LOCATION		LOP	LOCATION	
	DEGREES	MINUTES		DEGREES	MINUTES
22.00	147	10	35.00	156	51
23.00	147	51	36.00	157	41
24.00	148	33	37.00	158	32
25.00	149	15	38.00	159	24
26.00	149	58	39.00	160	17
27.00	150	41	40.00	161	12
28.00	151	25	41.00	162	8
29.00	152	9	42.00	163	5
30.00	152	54	43.00	164	3
31.00	153	40	44.00	165	3
32.00	154	27	45.00	166	5
33.00	155	14	46.00	167	9
34.00	156	2	47.00	168	14

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TABLE XXIII (Cont'd)
90° HCA DOUBLE TURN
FINAL TURN LEAD DISTANCE (DT-FLD) SCALE LOCATION

LOP	LOCATION	
	DEGREES	MINUTES
48.00	169	22
49.00	170	31
50.00	171	43
51.00	172	57
52.00	174	14
53.00	175	34
54.00	176	57
55.00	178	24
56.00	179	53
57.00	181	27
58.00	183	5
59.00	184	47

LOP	LOCATION	
	DEGREES	MINUTES
60.00	186	35
61.00	188	27
62.00	190	26
63.00	192	31
64.00	194	43
65.00	197	2
66.00	199	31
67.00	202	8
68.00	204	57
69.00	207	57
70.00	211	11

3.4.3.1.5.4 110° HCA, single turn scale. The 110° HCA SINGLE TURN scale shall be located within a circular band defined by the lower one-third of the 110° HCA section of the lead distance window as defined in 3.4.2.8 and shall be contained within the boundaries of the "SINGLE TURN" designation adjacent to this window following assembly of the computer. This scale shall contain graduation marks for each value of LOP angle between 18° and 52°. The location of these graduation marks shall be referenced clockwise from the Target TAS scale index and shall be positioned to within ± 10 arc minutes of the values specified in Table XXIV. These graduation marks shall be approximately 0.005 inches wide and 0.050 inches in length. Numerical designations shall be provided for every even value of LOP, starting with 18°. These numerical designations shall be approximately 7 point, medium face numerals of an appropriate style.

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TABLE XXIV
110° HCA
SINGLE TURN LEAD SCALE LOCATION

LOP	LOCATION	
	DEGREES	MINUTES
18	344	21
19	346	44
20	349	10
21	351	38
22	354	7
23	356	39
24	359	14
25	1	51
26	4	31
27	7	14
28	10	1
29	12	50
30	15	44
31	18	42
32	21	44
33	24	50
34	28	2
35	31	19
36	34	41
37	38	10
38	41	45

LOP	LOCATION	
	DEGREES	MINUTES
39	45	28
40	49	18
41	53	16
42	57	24
43	61	41
44	66	9
45	70	48
46	75	40
47	80	46
48	86	8
49	91	46
50	97	43
51	104	1
52	110	42

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3.4.3.1.5.5 110° HCA, Double Turn Factor (DTF) scale. The 110° DTF scale shall be located within a circular band defined by the center one-third of the 110° HCA section of the Lead Distance Window defined in 3.4.2.8 and shall be contained within the boundaries of the DTF designation adjacent to this window following assembly of the computer. This scale shall contain graduation marks for each value of LOP angle between 10° and 70°. The location of these graduation marks shall be referenced clockwise from the Target TAS scale index and shall be positioned within ± 10 arc minutes of the values specified in Table XXV. These graduation marks shall be approximately 0.005 inches wide and 0.050 inches in length. Numerical designations shall be provided for each value of LOP. These numerical designations shall be approximately 7 point, medium face numerals of an appropriate style.

TABLE XXV
110° HCA
DOUBLE TURN FACTOR SCALE LOCATION

LOP	LOCATION	
	DEGREES	MINUTES
9	49	21
10	33	18
11	18	47
12	5	31
13	353	20
14	342	2
15	331	30
16	321	38
17	312	20
18	303	34
19	295	14
20	287	19
21	279	45
22	272	31
23	265	34
24	258	53
25	252	26
26	246	13
27	240	11
28	234	20
29	228	39
30	223	7
31	217	44
32	212	28
33	207	20
34	202	18

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TABLE XXV (Cont'd)
110 HCA
DOUBLE TURN FACTOR SCALE LOCATION

LOP	LOCATION		LOP	LOCATION	
	DEGREES	MINUTES		DEGREES	MINUTES
35	197	21	56	105	29
36	192	31	57	101	16
37	187	45	58	97	2
38	183	4	59	92	46
39	178	27	60	88	29
40	173	53	61	84	9
41	169	23	62	79	48
42	164	57	63	75	24
43	160	33	64	70	57
44	156	11	65	66	28
45	151	52	66	61	54
46	147	35	67	57	17
47	143	19	68	52	36
48	139	5	69	47	50
49	134	52	70	42	59
50	130	39			
51	126	27			
52	122	16			
53	118	5			
54	113	53			
55	109	42			

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3.4.3.1.5.6 110° HCA, final lead - 30° turn scale. The 110° HCA, FINAL LEAD - 30° turn scale shall be located within a circular band defined by the upper one-third of the 110° HCA section of the lead distance window as defined in 3.4.2.8 and shall be contained within the boundaries of the "FINAL LEAD - 30° TURN" designation adjacent to the window following assembly of the computer. This scale shall contain graduation marks for each value of LOP angle between 18° and 52°. The location of these graduation marks shall be referenced clockwise from the Target TAS scale index and shall be positioned within ± 10 arc minutes of the values specified in Table XXVI. These graduation marks shall be approximately 0.005 inches wide and 0.050 inches in length. Numerical designations shall be provided for every fifth value of LOP between 20° and 40° and for every even value of LOP between 42° and 52°. These numerical designations shall be approximately 7 point, medium face numerals of an appropriate style.

TABLE XXVI
110° HCA FINAL LEAD - 30° TURN

LOP	LOCATION	
	DEGREES	MINUTES
18.00	83	50
19.00	85	36
20.00	87	23
21.00	89	13
22.00	91	4
23.00	92	58
24.00	94	54
25.00	96	52
26.00	98	54
27.00	100	58
28.00	103	5
29.00	105	15
30.00	107	29
31.00	109	46

LOP	LOCATION	
	DEGREES	MINUTES
32.00	112	7
33.00	114	33
34.00	117	3
35.00	119	38
36.00	122	18
37.00	125	4
38.00	127	55
39.00	130	54
40.00	133	59
41.00	137	12
42.00	140	33
43.00	144	3
44.00	147	43
45.00	151	34

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TABLE XXVI (Cont'd)
110° HCA FINAL LEAD - 30° TURN

LOP	LOCATION	
	DEGREES	MINUTES
46.00	155	36
47.00	159	52
48.00	164	22
49.00	169	7
50.00	174	10
51.00	179	33
52.00	185	17

3.4.3.2 Center disc, rear face. The rear face of the center disc shall contain a grid, a compass rose, a compressibility table and other special markings. The rear face of the center disk shall be in accordance with figure 3 and as defined in 3.4.3.2.1 through 3.4.3.2.3.

3.4.3.2.1 Grid. The rear face of the center disc shall contain a grid as shown in figure 3 and as defined in 3.4.3.2.1.1 through 3.4.3.2.1.6.

3.4.3.2.1.1 Grid reference. The grid reference shall consist of a 0.015 inch wide straight line passing through the center of the disk. This line shall extend 3.75 ± 0.025 inches from the center of the disk in one direction and 1.25 ± 0.020 inches from the center of the disk in the other direction. This line shall provide a reference for the grid as well as all other markings on the rear face of the center disk.

3.4.3.2.1.2 Left and right grid boundaries. The left and right boundaries of the grid shall be 0.020 inch wide lines located 3.000 ± 0.025 inches from, and parallel to, the grid reference line defined in 3.4.3.2.1.1. The left and right boundaries shall be terminated at the intersection of the upper and lower boundaries defined in 3.4.3.2.1.3 and 3.4.3.2.1.4.

3.4.3.2.1.3 Upper grid boundary. The upper grid boundary shall consist of a 0.015 inch wide circular arc, of radius 3.75 ± 0.025 inches. The center of this arc shall be located at the center of the disk. This arc shall pass through one end of the grid reference line and shall terminate at the intersection of the left and right boundaries defined in 3.4.3.2.1.2.

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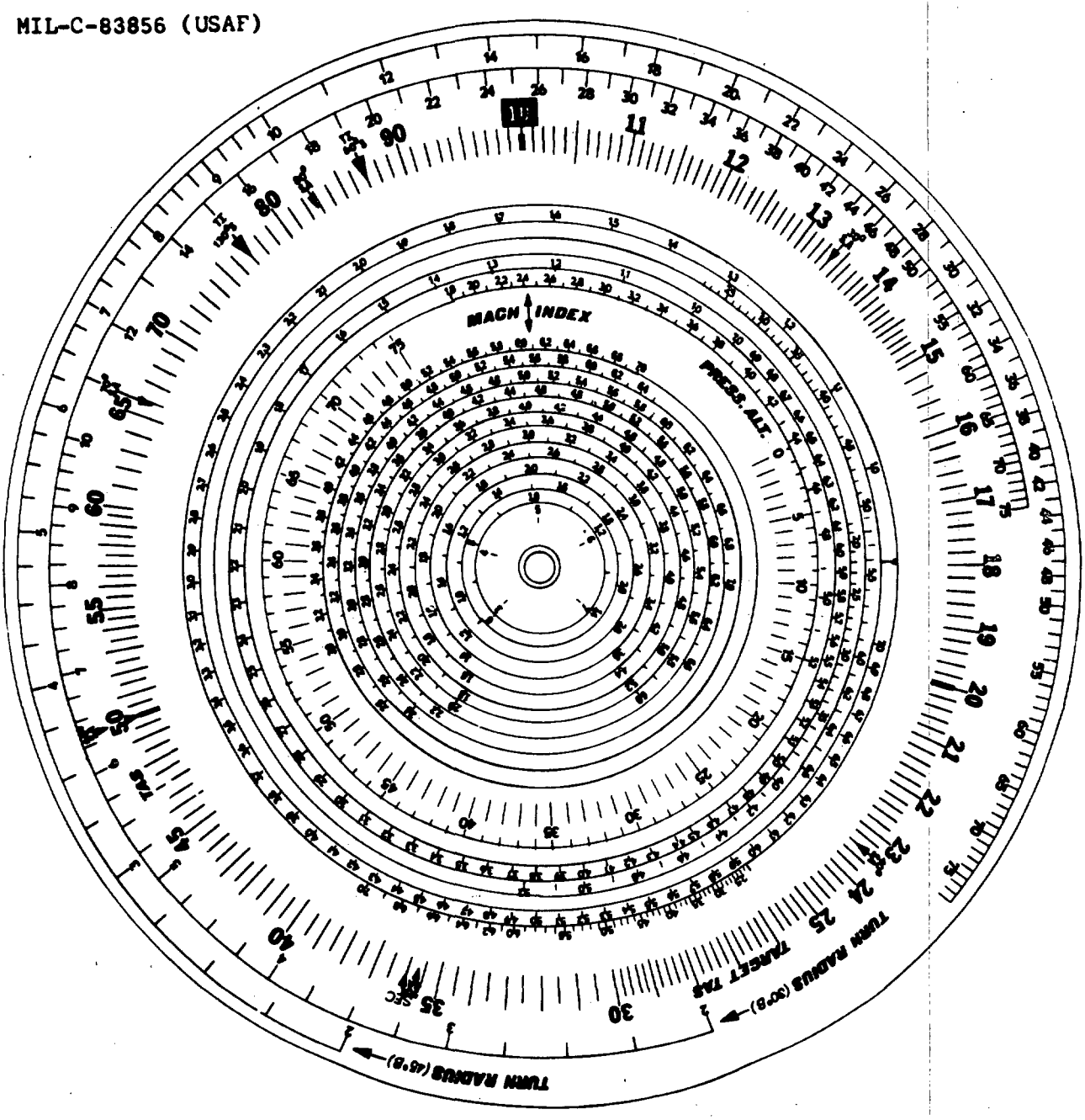


FIGURE 2. CENTER DISC, FRONT FACE

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3.4.3.2.1.4 Lower grid boundary. The lower grid boundary shall consist of a 0.015 inch wide line which is perpendicular to the grid reference line. The lower boundary shall pass through one end of the grid reference line and shall terminate at the intersection of the right and left boundaries defined in 3.4.3.2.1.2.

3.4.3.2.1.5 Primary grid divisions. The primary grid divisions shall consist of a 0.250 inch grid pattern. This grid pattern shall consist of 0.015 inch wide straight lines parallel to the grid reference line in one direction and perpendicular to the grid reference in the other direction. The perpendicular grid shall begin at the center of the disk and shall be terminated at the intersection of either the left and right boundaries or the circular arc of the upper boundary. The parallel grid shall begin at the reference line and shall terminate at the intersection of the upper and lower boundaries. The grid shall be constructed such that the appropriate dimension, referenced from either the grid reference line or the center of the disk, shall be accurate within + 0.020 inches. The tenth perpendicular grid line measured from the center of the disk along the grid reference toward the upper boundary shall be approximately 0.025 inches wide to provide a perpendicular grid reference.

3.4.3.2.1.6 Secondary grid divisions. The secondary grid divisions shall be located on the computer in accordance with 3.4.3.2.1.6.1 through 3.4.3.2.1.6.3.

3.4.3.2.1.6.1 Intermediate parallel grid divisions. Intermediate grid lines, parallel to the grid reference, shall be located one half way (0.125 inches) between each primary parallel grid line. These intermediate lines shall be approximately 0.010 inches wide and shall be clearly distinguishable from the primary grid. The intermediate parallel grid shall terminate at the upper and lower grid boundary.

3.4.3.2.1.6.2 Grid reference intermediate divisions. Intermediate divisions shall be placed along the grid reference as shown in Figure 3. These intermediate divisions shall consist of lines approximately 0.010 inch wide, perpendicular to the grid reference line. These lines shall sub-divide the grid reference into 0.050 inch increments and shall be positioned such as to equally sub-divide each primary perpendicular division along the grid reference into 5 equal divisions. Each of these intermediate divisions shall terminate at the first intermediate parallel grid to the right or left of the grid reference.

3.4.3.2.1.6.3 Lower intermediate perpendicular divisions. In addition to the divisions described in 3.4.3.2.1.6.2, intermediate perpendicular divisions constructed in the same manner as 3.4.3.2.1.6.2 shall be positioned in the lower area of the grid as shown in Figure 3. These divisions shall extend to the left and right of the reference line and terminate at the intersection of the fifth (5th) primary parallel grid line (1.250 inches from the grid reference line).

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3.4.3.2.2 Compass rose. The rear face of the center disc shall contain a compass rose as shown in Figure 3 and as defined in 3.4.3.2.2.1 through 3.4.3.2.2.2.

3.4.3.2.2.1 Compass rose major graduation marks. Major graduation marks for the compass rose shall be placed around the entire circumference of the rear face of the center disc in 10° ± 0.25° increments starting from the center of the grid reference line. Numerical designation and location of the major graduation marks shall be in accordance with Figure 3 of an appropriate size and style. The major graduation marks shall be so located such that the angular location of any mark shall be accurate to within 0.25° of the appropriate spacing from any other major graduation mark.

3.4.3.2.2.2 Compass rose minor graduation marks. A minor graduation mark shall be located one-half way between each major graduation mark. Numerical designation of minor graduation marks is not required. The minor graduation marks shall be located such that the angular location of any minor graduation mark shall be accurate to within ± 0.25° of the appropriate spacing from any other graduation mark.

3.4.3.2.3 Compressibility correction factor table. The rear face of the center computer disc shall contain a compressibility factor table to provide information necessary in the conversion of calibrated airspeed to true airspeed. This table shall be in accordance with Figure 3 and as defined in 3.4.3.2.3.1 through 3.4.3.2.3.3.

3.4.3.2.3.1 Compressibility correction factor data. The compressibility correction factor table, located on the rear face of the center disc of the computer shall contain correction factors for altitudes from 35,000 to 75,000 feet in 5,000 feet increments for calibrated airspeeds from 150 to 725 knots. Correction factors shall be provided for each 25 knot increment of calibrated airspeed. The data to be included in this table is contained in Table XXVII.

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TABLE XXVII - COMPRESSIBILITY CORRECTION FACTOR DATA

PRESSURE ALTITUDE FEET	CALIBRATED AIRSPEED - KNOTS															
	150	175	200	225	250	275	300	325	350	375	400	425	450	475	500	525
35,000	3	5	7	9	12	16	20	24	29	35	40	45	50	55	60	63
40,000	4	6	9	12	15	20	25	31	37	43	48	54	59	64	68	72
45,000	5	8	11	15	20	25	32	38	44	50	56	61	66	71	75	78
50,000	7	10	14	19	25	31	38	45	51	58	63	68	73	77	81	84
55,000	8	12	17	23	30	37	44	50	57	63	69	74	79			

PRESSURE ALTITUDE	150	175	200	225	250	275	300	325	350	375	400	425
60,000	11	16	21	28	35	42	50	56	62	68	73	77
65,000	13	19	26	33	40	47	54	60	66	72		
70,000	16	23	29	37	44	51	58	64	69			
75,000	19	26	33	40	47	55	61					

PRESSURE ALTITUDE	550	575	600	625	650	675	700	725
35,000	67	70	72	74	76	76	76	75
40,000	75	78	80	81	82			
45,000	82	84						

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3.4.3.2.3.2 Dimensions and layout. The compressibility correction factor table shall be arranged and positioned as shown in figure 3. The boundary of the table shall consist of lines approximately 0.005 inches wide. The upper horizontal boundary shall be coincident with the lower grid boundary of 3.4.3.2.1.4. The left and right perpendicular boundaries as well as the location of data within the table shall be determined by the extension of the major grid lines of 3.4.3.2.1.5. The lower boundary of the table shall be located at approximately 3.100 inches from the center of the disc.

3.4.3.2.3.3 Table designation. The table shall be designated as shown in figure 3 using the following approximate letter sizes of an appropriate style.

a. CALIBRATED AIRSPEED - KNOTS in 7 point Gothic.

b. PRESSURE ALTITUDE FEET, 4 point Gothic.

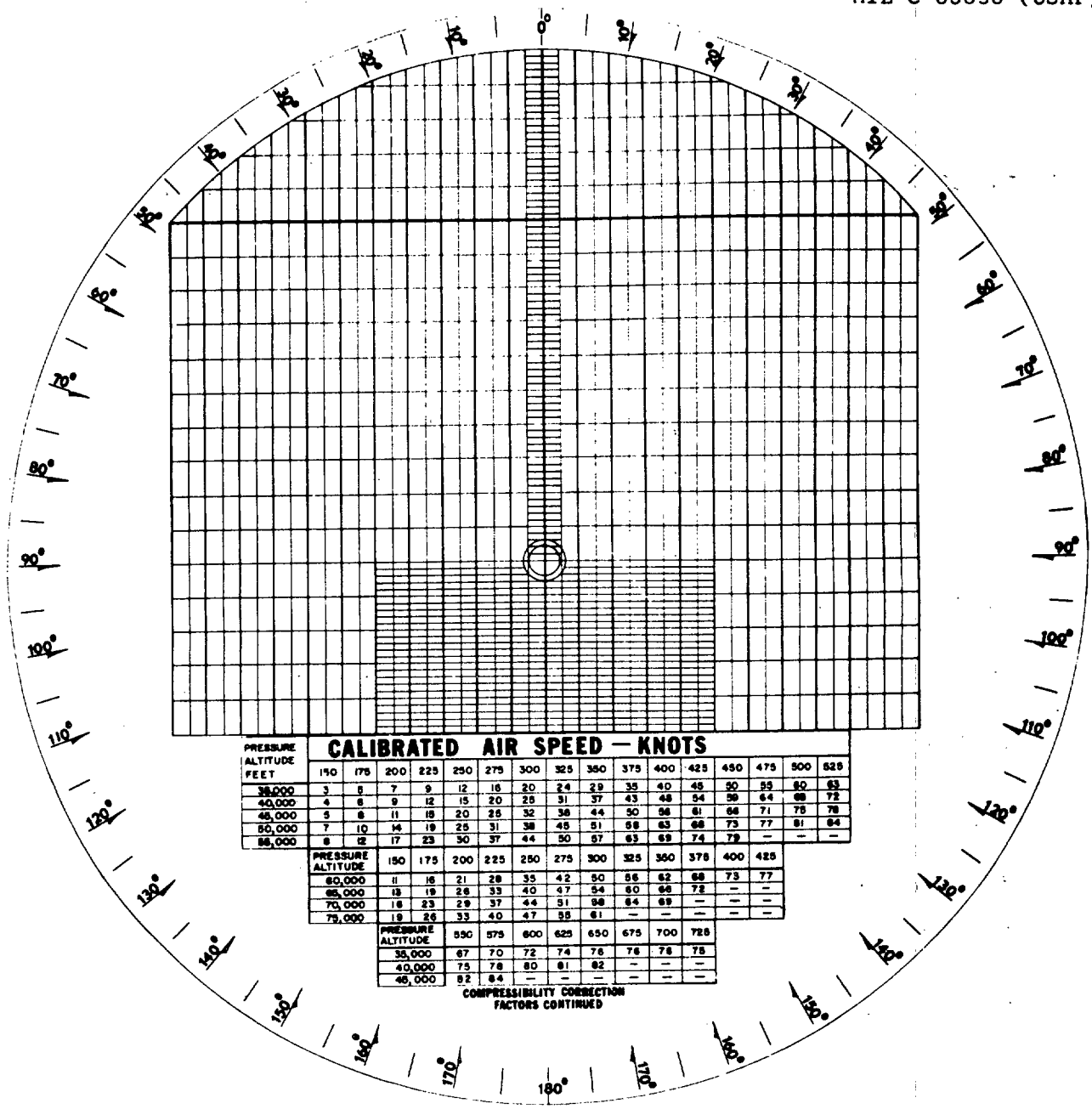
c. "COMPRESSIBILITY CORRECTION FACTORS CONTINUED" in 4 point Gothic.

d. The numerical designations for values of pressure altitude, calibrated airspeed, and correction factors shall be 4 point, bold face Gothic numerals.

3.4.4 Rear computer disc. The rear computer disc shall be of a clear transparent material in accordance with Federal Specification L-P-535, Composition B, Type II, Grade C and shall contain a compass rose. This disc shall be 4.500 ± 0.025 inches in radius. The layout and arrangement of the compass rose on this disc shall be in accordance with figure 4 and as defined in 3.4.4.1.

3.4.4.1 Rear computer disc compass rose. The rear computer disc shall contain graduation marks for each one degree increment of arc from 0° to 359°. The graduation mark for each 5° increment shall be 0.250 ± 0.010 inches in length and each intermediate graduation mark shall be 0.125 ± 0.010 inches in length. Each graduation mark shall extend radially outward from a distance of 3.375 ± 0.010 inches from the center of the disc. Appropriate numerical designations shall be provided above each 10° increment of arc. These numerical designations shall be of 7 point Gothic numerals of an appropriate style. The graduation marks for this compass rose shall be positioned such that an angular accuracy of ± 20 arc minutes will exist between any two graduation marks on the compass rose.

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CALIBRATED AIR SPEED — KNOTS

PRESSURE ALTITUDE FEET	150	175	200	225	250	275	300	325	350	375	400	425	450	475	500	525
35,000	3	5	7	9	12	16	20	24	29	35	40	46	50	55	60	65
40,000	4	6	9	12	15	20	25	31	37	43	48	54	59	64	68	72
45,000	5	8	11	15	20	25	32	38	44	50	56	61	66	71	75	78
50,000	7	10	14	19	25	31	38	45	51	58	63	68	73	77	81	84
55,000	8	12	17	23	30	37	44	50	57	63	69	74	79	—	—	—

PRESSURE ALTITUDE	150	175	200	225	250	275	300	325	350	375	400	425
60,000	11	16	21	28	35	42	50	56	62	68	73	77
65,000	13	19	26	33	40	47	54	60	66	72	—	—
70,000	16	23	29	37	44	51	58	64	69	—	—	—
75,000	19	26	33	40	47	55	61	—	—	—	—	—

PRESSURE ALTITUDE	550	575	600	625	650	675	700	725
35,000	67	70	72	74	76	76	76	76
40,000	75	78	80	81	82	—	—	—
45,000	82	84	—	—	—	—	—	—

COMPRESSIBILITY CORRECTION FACTORS CONTINUED

FIGURE 3. CENTER DISC, REAR FACE

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TABLE XXVIII
DT FACTORS

ITAS	DT+
350	1.0
400	1.5
450	1.5
500	2.0
550	2.0
600	2.0
650	2.0
700	2.5
750	2.5
800	3.0
850	3.0
900	3.0

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection and testing of the computer shall be classified as follows:

a. First Article Inspection (see 4.4)

b. Quality Conformance Inspection (see 4.5)

4.3 Test conditions

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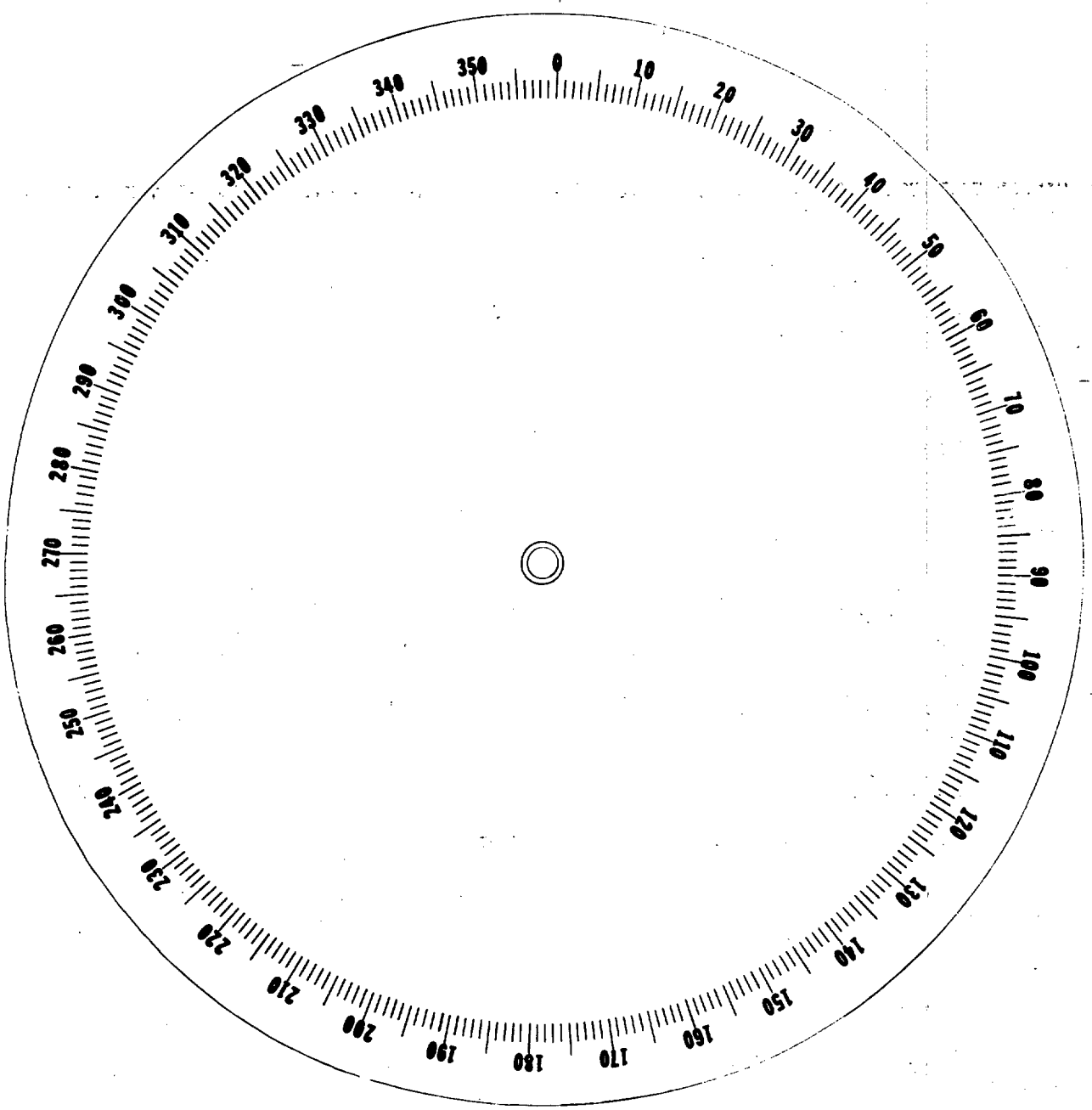


FIGURE 4. REAR DISC

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4.3.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 25 ± 5 degrees C.

4.4 First article inspection

4.4.1 First article test sample. The test sample shall consist of three assembled computers representative of the production equipment. The sample shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.1.1 Data to accompany test sample.

a. Operating instructions to correctly operate the computer.

b. Certification that all materials used in the production of the computers are in accordance with the requirements of this specification.

4.4.2 Test report, test sample and data for the procuring activity. When the tests are conducted at a location other than the laboratory of the procuring activity, the following shall be furnished to that activity.

a. Test report. Three copies of a test report in accordance with MIL-STD-831.

b. Test sample. The sample which was tested and an untested sample.

c. Data. The data required under paragraph 4.4.1.1.

4.4.3 First article tests. The first article tests shall consist of the acceptance tests of 4.5 and all of the functional accuracy test specified in 4.4.4.1 through 4.4.4.7.

4.4.4 FUNCTIONAL ACCURACY TESTS.

4.4.4.1 Interceptor and Target True Airspeed Accuracy Tests. In each of the following tests, set the interceptor true airspeed index over the centerline of the specified value on the target true airspeed scale. The value indicated on the target true airspeed scale opposite the specified value on the interceptor true airspeed scale shall be within the limits indicated in Table XXIX.

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TABLE XXIX
I TAS AND T TAS TEST

SET I TAS TO:	READ VALUE ON T TAS OPPOSITE:	VALUE ON T TAS SHALL BE: Minimum	Maximum
13.40	23.80	31.80	32.00
	42.00	56.20	56.40
	75.00	10.03	10.07
12.00	21.60	25.84	25.98
	38.50	46.10	46.30
	68.00	81.50	81.70
16.00	29.00	46.30	46.50
	51.00	81.40	81.80
	91.00	14.52	14.60

4.4.4.2 30° and 45° Turn radius scale accuracy tests. In each of the following tests, set the interceptor true airspeed index over the centerline of the graduation mark for the value of turn radius specified. The value indicated on the target true airspeed scale under the index shall be within the limits indicated in Tables XXX and XXI.

4.4.4.2.1 30° Turn radius scale accuracy. Use Table XXX.

TABLE XXX
30° TURN RADIUS SCALE TEST

Turn Radius	TARGET TAS INDICATION Minimum	Maximum
2.75	329.0	331.0
4.00	397.0	399.0
6.75	516.0	517.0
11.00	659.0	661.0
20.00	889.0	891.0
30.00	1089.0	1091.0
44.00	1319.0	1321.0
73.00	1699.0	1701.0

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4.4.4.2.2 45° Turn radius scale accuracy. Use Table XXXI.

TABLE XXXI
45° TURN RADIUS SCALE TEST

TURN RADIUS	TARGET TAS INDICATION	
	Minimum	Maximum
3.50	489.0	491.0
5.25	599.0	601.0
7.25	704.0	706.0
14.00	979.0	981.0
21.00	1199.0	1201.0
29.00	1409.0	1411.0
51.00	1869.0	1871.0
75.00	2190.0	2192.0

4.4.4.3 Temperature-Altitude window scale accuracy tests.

4.4.4.3.1 Front disc-temperature scale accuracy. In each of the following tests, set the specified value of temperature to the 10,000 feet graduation mark on the center disc pressure altitude scale. The value indicated on the target true airspeed scale under the interceptor true airspeed scale index shall be within the limits indicated in Table XXXII.

TABLE XXXII
FRONT DISC-TEMPERATURE
SCALE ACCURACY TEST

TEMPERATURE °C	TARGET TAS INDICATION	
	Minimum	Maximum
+40	12.55	12.60
+30	12.35	12.40
+15	12.02	12.08
0	11.72	11.76
-20	11.30	11.35
-30	11.05	11.10
-40	10.82	10.86
-50	10.59	10.63
-60	10.35	10.39
-70	10.10	10.14

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4.4.4.3.2 Front disc, standard day scale accuracy tests. In each of the following tests, set the specified value of "standard day pressure altitude" on the front disc to the 10,000 feet graduation mark on the center disc pressure altitude scale. The value indicated on the target true airspeed scale under the intercept true airspeed index shall be within the limits indicated in Table XXXIII.

TABLE XXXIII
FRONT DISC, STANDARD
DAY SCALE ACCURACY TEST

STANDARD DAY PRESSURE ALTITUDE Thousands of Feet	TARGET TAS INDICATION	
	Minimum	Maximum
0	12.02	12.08
10	11.61	11.65
20	11.18	11.22
30	10.73	10.77
35	10.48	10.52

4.4.4.3.3 Center disc-pressure altitude scale accuracy tests. In each of the following tests, set the zero (0) standard day pressure altitude graduation mark on the front disc to the value specified on the center disc pressure altitude scale. The value indicated on the target true airspeed scale under the interceptor true airspeed index shall be within the limits indicated in Table XXXIV.

TABLE XXXIV
CENTER DISC-PRESSURE ALTITUDE
SCALE ACCURACY TEST

PRESSURE ALTITUDE	TARGET TAS INDICATION	
	Minimum	Maximum
0	9.98	10.02
5	10.94	10.98
10	12.04	12.08
15	13.00	13.04
20	14.72	14.78
25	16.38	16.46
30	18.50	18.60
35	20.55	20.65
45	26.15	26.25
55	33.20	33.40
65	42.30	42.50
75	53.70	53.90

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4.4.4.3.4 Mach index position accuracy test. Set the -15° graduation mark on the temperature-altitude window opposite the "MACH-INDEX" on the center disc pressure altitude scale. The value indicated on the target true airspeed scale under the interceptor true airspeed index shall be between 66.10 and 66.20.

4.4.4.4 LOP window-LOP scale accuracy tests. In each of the following tests, set the interceptor true airspeed index to the value of LOP indicated for each beam attack angle specified. The value on the target true airspeed scale under the interceptor true airspeed index shall be within the limits indicated in Table XXXV.

TABLE XXXV
LOP WINDOW-LOP SCALE
ACCURACY TEST

BEAM ATTACK ANGLE	LOP	TARGET TAS SCALE INDICATION	
		Minimum	Maximum
60°	30°	0.499	0.501
	60°	0.998	1.002
70°	45°	0.779	0.781
	55°	0.998	1.002
80°	46°	0.888	0.890
	50°	0.998	1.002
90°	31°	0.600	0.602
	35°	0.699	0.701
	41°	0.868	0.870
	45°	0.998	1.002
	49°	1.149	1.151
	51°	1.233	1.237
	53°	1.325	1.330
	57°	1.538	1.542
	61°	1.802	1.806
100°	20°	0.394	0.396
	40°	0.998	1.002
110°	19°	0.418	0.420
	23°	0.533	0.535
	27°	0.665	0.667
	29°	0.738	0.740
	33°	0.904	0.906
	35°	0.998	1.002
	37°	1.104	1.106
	39°	1.220	1.224
	49°	2.100	2.110

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TABLE XXXV (Cont'd)
LOP WINDOW-LOP SCALE
ACCURACY TEST

BEAM ATTACK ANGLE	LOP	TARGET TAS SCALE INDICATION	
		Minimum	Maximum
120°	28°	0.885	0.887
	30°	0.998	1.002
130°	21°	0.738	0.740
	25°	0.998	1.002
140°	18°	0.824	0.826
	20°	0.998	1.002
150°	11°	0.585	0.587
	15°	0.998	1.002
160°	6°	0.431	0.433
	10°	0.998	1.002
170°	5°	0.998	1.002

4.4.4.5 Lead distance window, lead distance scale accuracies. In each of the following tests, set the interceptor true airspeed index over the centerline of the specified graduation mark within the lead distance window. The value indicated on the target true airspeed scale under the interceptor true airspeed index shall be within the limits specified in Table XXXVI.

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TABLE XXXVI

LEAD DISTANCE WINDOW LEAD DISTANCE SCALE ACCURACY TEST

Lead Distance Scale	Graduation Mark	Target TAS Indication	
		Minimum	Maximum
90° HCA Single Turn	22 33 44 55 66	1.2290 1.369 1.545 1.820 2.276	1.232 1.372 1.555 1.830 2.284
90° HCA D.T.F.	10 25 40 55 70	.016150 .006690 .004190 .002890 .001895	.016250 .006710 .004210 .002910 .001905
90° HCA FINAL LEAD-40° Turn	22 34 46 58 70	.2560 .2710 .2910 .3220 .3855	.2568 .2718 .2918 .3230 .3865
110° HCA Single Turn	18 26 35 43 52	.904 1.028 1.220 1.482 2.025	.906 1.030 1.224 1.486 2.035
110° HCA D.T.F	10 25 40 55 70	.0123500 .004980 .002990 .001990 .001295	.012450 .005020 .003010 .002010 .001305
110° HCA FINAL LEAD-D.T.	18 26 35 43 52	.1706 .1880 .2146 .2510 .3168	.1714 .1888 .2154 .2518 .3176

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4.4.4.6 Compressibility table data accuracy tests. The data contained within the compressibility tables on the front disc and rear face, the center disc shall be visually inspected to verify that this data is in accordance with tables IV and XXVII respectively.

4.4.4.7 Compass rose-attack indices accuracy test. In each of the following tests the centerline of the angular value specified for the compass rose (rear disc) shall be placed over the centerline of the vertical grid reference line on the rear face, the center disc. Alignment between the graduation marks on the compass rose of the rear computer disc and the compass rose at the rear face of the center disc (Attack indices) shall be as specified in the table XXXVIII.

4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests.

b. Sampling tests.

4.5.1 Individual tests. Each computer shall be subjected to the following tests.

4.5.1.1 Examination of product. Each computer shall be examined to determine the compliance with this specification with respect to material and workmanship.

4.5.1.2 Centering and scale alignment test. The following tests shall be performed on each computer submitted for acceptance.

4.5.1.2.1 Front-Center disc centering tests. The centering of the front computer disc with respect to the center computer disc shall be performed as follows:

a. Align the interceptor true airspeed index with the centerline of the target true airspeed index. The graduation marks for "20" and "50" on the interceptor true airspeed scale shall align with the graduation marks for the same respective values on the target true airspeed scale, within the limits of the special graduation marks provided for this test. (See 3.4.3.1.1.2)

b. Align the interceptor true airspeed index with the centerline of the graduation mark for "20" on the target true airspeed scale. The graduation marks for "25" and "50" on the interceptor true airspeed scale shall align with the graduation marks for "50" and "10" respectively on the target true airspeed scale within the limits defined by the special graduation marks provided for this test. (See 3.4.3.1.1.2)

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c. Align the interceptor true airspeed index with the centerline of the graduation mark for "50" on the target true airspeed scale. The graduation marks for "20" and "40" on the interceptor true airspeed scale shall align with the graduation marks for "10" and "20" respectively on the target TAS scale within the limits defined by the special graduation marks provided for this test (See 3.4.3.1.1.2).

4.5.1.2.2 Rear-Center disc centering tests. With the "0°" graduation mark on the compass rose (rear disc) aligned with the centerline of the vertical grid reference line on the rear face of the center disc, compass rose graduation marks for 90°, 180° and 270° shall align with the attack indices graduation marks for 90° CW, 180° and 90° CCW, respectively.

4.5.1.3 Scale alignment test. The scale alignment between various scales contained on the front face of the center computer disc and the front computer disc shall be verified on each computer submitted for acceptance. This alignment shall be verified by aligning the interceptor true airspeed index with the centerline of the target true airspeed index and verifying that alignment exists between the various other graduation marks listed in tables below, within the tolerance specified in Table XXXVII.

TABLE XXXVII
SCALE ALIGNMENT TEST

1st GRADUATION MARK	2nd GRADUATION MARK	REQUIRED ALIGNMENT
Interceptor TAS Index	60° LOP on Beam Attack Scale-LOP Window	2 Line Widths
Interceptor TAS Index	55° LOP on 70° Beam Attack Scale-LOP Window	
	50 80 45 90 40 100 35 110 30 120 25 130 20 140 15 150	
ITAS	10 160 5 170	2 Line Widths

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TABLE XXXVII (Cont'd)
SCALE ALIGNMENT TEST

"0" on Pressure	"0" on "ALT" Scale Front Disc, Temp-Alt Window	2 Line Widths
"0" on Pressure Altitude Scale Center Disc	+15°C on Temp Scale, Front Disc, Temp-Alt Window	2 Line Widths

4.5.2 Sampling plan and tests.

4.5.2.1 Sampling plan. One computer shall be selected at random from every 100 computers or fraction thereof produced and subjected to the Functional Accuracy Tests defined in 4.4.4.

4.5.2.2 Rejection and retest. When a computer subjected to the sampling tests fails to meet the requirements of this specification, computers still on hand or produced subsequent to the failed unit shall not be accepted on the basis of individual tests until the extent and cause of failure are determined. However, at the option of the vendor, computers produced prior to the failed unit may be submitted for acceptance under the provisions that all such units be subjected to and successfully meet the requirements which constituted rejection of the failed unit as well as the individual test requirements.

4.5.2.3 Individual tests may continue. At the option of the vendor, individual tests may be continued pending the investigation of a sampling plan test failure. However, final acceptance of computers on hand or produced subsequent to the sampling plan failure shall not be accomplished until it is determined by the procuring activity that these computers meet the requirement of this specification.

4.5.2.4 Defects in computers already accepted. The investigation of a test failure could indicate that defects may exist in computers already accepted. If so, the contractor shall fully advise the procuring activity of all defects to be found and methods of correcting them.

4.6 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

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TABLE XXXVIII
COMPASS ROSE - ATTACK INDICES
ACCURACY TEST

ANGULAR VALUE	GRADUATION	
	Rear Disc	Attack Indices
0	20°	20 CW
	40°	40 CW
	60°	60 CW
	80°	80 CW
	100°	100 CW
	120°	120 CW
	140°	140 CW
	160°	160 CW
	200°	160 CCW
	220°	140 CCW
	240°	120 CCW
	260°	100 CCW
	280°	80 CCW
	300°	60 CCW
	320°	40 CCW
	340°	20 CCW
90	100°	10 CW
	120°	30 CW
	140°	50 CW
	160°	70 CW
	180°	90 CW
	200°	110 CW
	220°	130 CW
	240°	150 CW
	260°	170 CW
	280°	170 CCW
	300°	150 CCW
	320°	130 CCW
	340°	110 CCW
	0°	90 CCW
	20°	70 CCW
	40°	50 CCW
	60°	30 CCW
	80°	10 CCW
180	195°	15 CW
	215°	35 CW
	235°	55 CW
	255°	75 CW
	275°	95 CW
	295°	115 CW
	315°	135 CW

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TABLE XXXVIII
COMPASS ROSE - ATTACK INDICES
ACCURACY TEST
(Cont'd)

ANGULAR VALUE	GRADUATION	
	Rear Disc	Attack Indices
	335°	155 CW
	355°	175 CW
	15°	165 CCW
	35°	145 CCW
	55°	125 CCW
	75°	105 CCW
	95°	85 CCW
	115°	65 CCW
	135°	45 CCW
	155°	25 CCW
	175°	5 CCW
270	275°	5 CW
	295°	25 CW
	315°	45 CW
	335°	65 CW
	355°	85 CW
	15°	105 CW
	35°	125 CW
	55°	145 CW
	75°	165 CW
	95°	175 CCW
	115°	155 CCW
	135°	135 CCW
	155°	115 CCW
	175°	95 CCW
	195°	75 CCW
	215°	55 CCW
	235°	35 CCW
	255°	15 CCW

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Each computer to be preserved and packaged in quantity unit packs of one each in accordance with Method III of MIL-P-116. Computer in plastic case to be wrapped with L-P-378 and placed in a form fitting setup box in accordance with PPP-B-676 unless otherwise stipulated in the contract or order by the procuring activity.

5.2 Packing. Each computer shall be packed for shipment in accordance with Level B requirements as defined within MIL-STD-794, unless otherwise specified in the contract or order by the procuring activity.

6. NOTES

6.1 Intended use. The computer covered by this specification is intended for use in solving air navigation attack problems.

6.2 Ordering data. Procurement Documents should specify the following:

a. Title, number and date of this specification.

b. Level of preservation and packaging.

c. Whether or not preparation of operational manual is required.

Custodian:
Air Force - 71

Preparing Activity:
Air Force - 71

Project Number
6605-F163

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products *for ultimate use by the Department of Defense*. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

Air Force OC, 29 Apr 64-400

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